

Extreme Risk Analysis: U.S. Census Data

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1 Introduction

Understanding how city populations are distributed is key to urban economics and demography. City populations are not distributed uniformly across the United States. The vast majority of cities are small, yet a handful of large metropolitan areas, e.g. New York, Los Angeles and Chicago, dwarf the rest by orders of magnitude. In this context, this project aims to address the following question: **What is the extreme value behavior of population sizes in U.S. cities?** The analysis relies on the dataset *census2010.xlsx*, which provides population size information on large U.S. cities over the period 2010–2019. The dataset contains several variables describing the population size of each city, such as the rank of cities according to their population size, the official population report during the 2010 U.S. Census. The main variables used in the analysis are the annual population estimates from 2010 to 2019, denoted as **Population Estimate**. These variables provide a time series of population sizes and allow for the study of both the evolution of city populations over time and the spatial heterogeneity among the U.S cities.

The analysis is organised as follows. Section 2 presents the exploratory data analysis. Section 3 motivates the use of Extreme Value Theory. Section 4 examines the temporal stability of the tail index over the period 2010–2019. Section 5 investigates the spatial heterogeneity of extreme value behavior across the four US Census regions. Section 6 addresses the prediction question: given the estimated tail behavior, what is the expected distribution of city sizes in 2040? Finally, Section 7 concludes with a summary of the main findings.

2 Exploratory Data Analysis

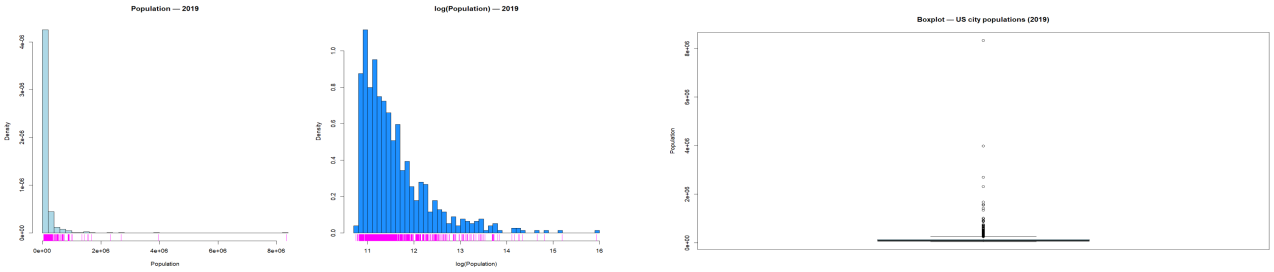
Firstly, our dataset contains a total of 788 cities with annual population estimates from 2010 to 2019. As shown in Tables 1 and 2, both the mean and median populations exhibit a steady upward trend over the decade, from 150,835 and 80,360 in 2010 to 162,618 and 85,686 in 2019, indicating overall urban growth. The minimum population also increases over time, from 22,520 in 2010 to 46,536 in 2019, suggesting that even the smallest cities are expanding. In contrast, the maximum population remains relatively stable, increasing by only 0.18% between 2010 and 2019. Furthermore, the mean is approximately twice the median across all years, indicating an evidence of a right-skewed distribution driven by a small number of very large cities constituting the upper tail. Across the pooled sample of 7,880 observations, approximately 95% of cities have a population **below 462,638** inhabitants, while only the top 1% **exceed 1,337,831**, and the maximum reaches **8,469,153**. This behavior follows a **"many small / few huge"** pattern typical of **heavy-tailed data**. This is clearly visible in Figure 1a, which displays the raw and log-transformed population distributions for 2019, the most recent year in our sample. For simplicity, only 2019 is displayed here, but the same pattern is consistently observed across all years from 2010 to 2019. The raw distribution is clearly right-skewed: the majority of cities have a population below 2 million inhabitants, while a small number of large cities such as New-York and Los Angeles reaching near 4 million of inhabitants and more. While the log-distribution produces a more bell-shaped distribution, it remains right-skewed distribution. This pattern is also clearly observable on the boxplot on the Figure 1b. The persistence of a heavy tail motivates the use of Extreme Value Theory (EVT), which is specifically designed to model the tail of a distribution rather than its bulk.

Table 1: Overall pooled summary statistics (2010–2019)

Sample	n	Min	Max	Mean	Median	SD	P95	P99
Pooled 2010–2019	7,880	22,520	8,469,153	157,379	83,098	382,181	462,638	1,337,831

Table 2: Annual summary statistics by year (populations)

Year	n	Min	Max	Mean	Median	SD	P95	P99
2010	788	22,520	8,190,209	150,835	80,360	371,186	435,552	1,214,072
2011	788	23,364	8,272,948	152,362	80,921	374,828	440,913	1,231,452
2012	788	24,378	8,346,693	154,005	81,948	378,482	447,626	1,254,421
2013	788	26,278	8,396,091	155,500	82,520	381,345	451,791	1,271,378
2014	788	28,461	8,433,806	157,008	82,784	383,889	456,852	1,291,673
2015	788	30,572	8,463,049	158,568	83,400	386,187	462,003	1,312,508
2016	788	35,049	8,469,153	159,965	84,106	387,570	465,481	1,334,078
2017	788	39,166	8,437,478	161,058	84,809	387,531	466,342	1,351,597
2018	788	44,663	8,390,081	161,865	85,200	386,689	471,428	1,352,217
2019	788	46,536	8,336,817	162,618	85,686	385,697	476,749	1,354,009



(a) Distribution of city populations (raw and log-scale)

(b) Boxplot of city populations

Figure 1: Distributional features of US city populations in 2019: (a) strong right-skewness visible in both raw and log scales, and (b) presence of extreme values highlighted by the boxplot.

As done in the practicals, to further confirm the heavy-tailed nature, Figure 2 presents the Moment estimator $\hat{\gamma}$ plotted against k, the number of top order statistics. We can observe that the moment estimator of the Extreme Value Index remains above zero over a large range of k values (1<k<=500). For simplicity, only the years 2010, 2014 and 2019 are displayed here, however the same pattern has been observed across the years from 2010 to 2019 (see Figure 11 in Appendix Section A.1). This provides a strong evidence that the Extreme Value Index, γ is greater than zero, indicating that the distribution of US city populations belongs to the **Fréchet domain of attraction** and is therefore **heavy-tailed**.

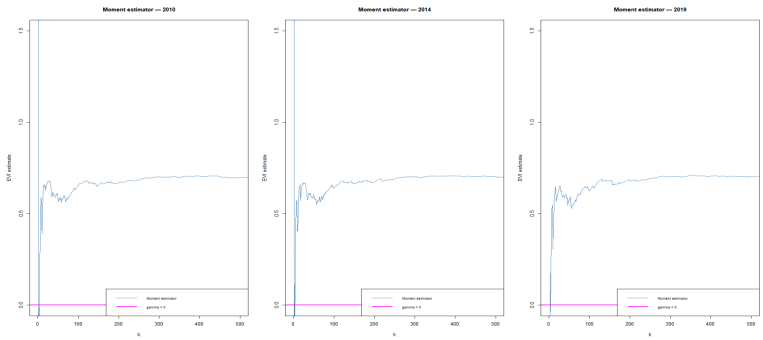
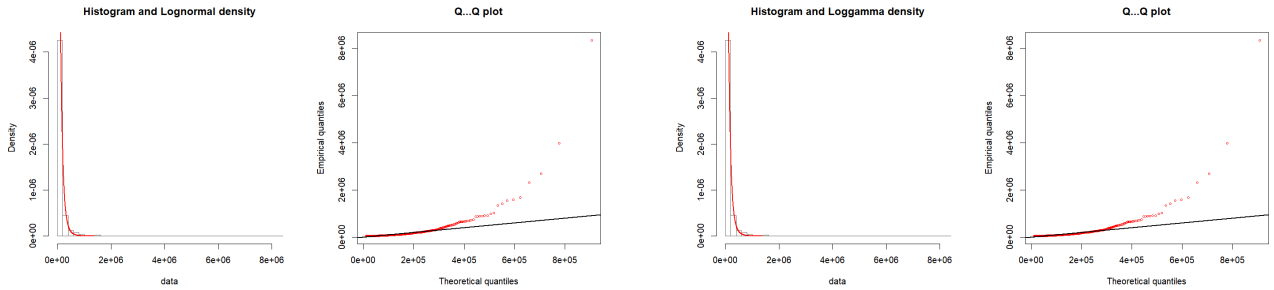


Figure 2: Moment estimator of the Extreme Value Index plotted against k, the number of top order statistics for 2010, 2014 and 2019.

3 Why Modelling the Bulk is Not Appropriate ?

Now, let's try to fit the data with a distribution, we started with a naive check trying the log-normal distribution on Figure 3a. We can observe on the QQ plot that the log-normal distribution **fits nicely the bulk but badly underestimates the extremes** (empirical quantiles > theoretical quantiles). We have found that the 99% log-normal theoretical quantile is approximately $553{,}360$ inhabitants, whereas the empirical 99% quantile reaches $1{,}354{,}009$ inhabitants. The lognormal distribution thus understates extreme city sizes by $\mathbf{800{,}650}$ inhabitants, a gap of $\mathbf{59.1}\%$. Conversely, the 90% theoretical quantile ($262{,}757$ inhabitants) is nearly identical to its empirical counterpart ($262{,}200$ inhabitants), with a gap of only $\mathbf{0.2}\%$. This confirms that the distribution seems to be a good fit for the bulk but not for the extreme values. In addition, we have fitted other distributions belonging to the Fréchet domain of attraction, as we saw previously that our data is heavy-tailed. As shown Figure 3b, the Log-gamma distribution fit demonstrates that also here the distribution succeed to fit the bulk but completely underestimate the extremes quantiles. Indeed the log-gamma distribution fits the bulk accurately (gap of only $\mathbf{0.3}\%$ at the quantile 90%) but severely underestimates the upper tail (gap of $\mathbf{59.1}\%$ at the quantile 99%), identical to the lognormal result.

Taken together, this illustrates the limitation of bulk modeling: fitting a parametric distribution to the entire dataset forces a tradeoff between the fit of the bulk or the fit of the tail. Since our study here focuses on the analysis of extreme values, Extreme Value Theory adresses this limitation by focuses only on the far right tail : modeling maxima and threshold exceedances to estimate extreme quantiles.



(a) Lognormal fit (naive approach)

(b) Log-gamma fit

Figure 3: Distributional fits to US city population sizes in 2019: histogram and QQ-plots.

4 Extreme Value Analysis: Temporal Stability of the Tail Index (2010–2019)

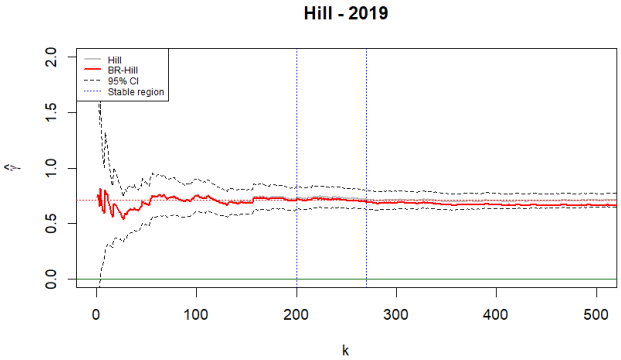
Before conducting the extreme value analysis, we address the question of the data structure. As noted in the course practicals, the theory underlying the Hill and Weissman estimators is derived for i.i.d. data. Rather than pooling all years together into a single sample which would introduce dependencies, we therefore treat each annual cross-section as an independent i.i.d. sample of $n = 788$ cities. Moreover, treating each year separately allows us to directly address the question of temporal stability: if the tail index $\hat{\gamma}$ estimated independently for each year remains stable over the decade.

4.1 Methodology: Tail Index Estimation on 2019 Data

In this section, we will focus to estimate the tail Index of city populations in 2019. Since we have shown previously that the data is heavy-tailed, we use the Hill and Bias-Reduced Hill estimators to find reasonable and refined estimates of the tail index. The **Hill (1975)** estimator is defined such as $k \to \infty$, $k/n \to 0$, and $t = X_{n-k,n}$, and use the top k exceedances to define the :

$$\hat{\gamma}_k = \frac{1}{k}\sum_{i=1}^{k} \log \frac{X_{n-i+1,n}}{X_{n-k,n}} = \frac{1}{k}\sum_{i=1}^{k} \log X_{n-i+1,n} - \log X_{n-k,n}$$

We plot both estimators against a range of values of k, identify the first stable region visually, and select k^* as the upper bound of this stable region, following the approach of *Pratical number 5 : PRC*. As Figure 4 (left) demonstrates, we manually select the stable region $[200, 270]$, delimited by the blue vertical lines, and take $k^* = 270$ as our optimal threshold. This yields a Hill estimate of $\hat{\gamma} = 0.7306$ and a Bias-Reduced Hill estimate of $\hat{\gamma} = 0.7118$. As a further check, we apply the automated dAMSE procedure, which selects k^* by minimising the asymptotic mean squared error of the Hill estimator, yielding $\hat{\gamma} = 0.7064$. The results are summarised in Table 4 (right). Overall, these findings confirm that the distribution of US city populations is heavy-tailed and therefore implying a tail index $\hat{\gamma}$ greater than 0. In addition, since $\hat{\gamma}$ is greater than 1/2 and lower than 1, the mean of the distribution is finite while the variance is infinite, reflecting the extreme heterogeneity in US city sizes.



Year	Stable k	Hill	BR-Hill	dAMSE
2019	[200, 270]	0.7306	0.7118	0.7064

Figure 4: Extreme value analysis for US city populations in 2019: Hill plot of the tail index $\hat{\gamma}$ as a function of k (left), and corresponding estimates using Hill, bias-reduced Hill, and dAMSE methods (right).

Once we have used the Hill estimator to get the estimation of the Extreme Value Index, $\hat{\gamma}$, the next step is to check if k has been chosen correctly and if the estimate $\hat{\gamma}$ of γ is reasonable. To check it we are relying on the **Exponential QQ-plot** and **Pareto QQ-plot** displayed in the Figure 32 in Appendix Section A.2. Both graphs demonstrate that the points are closely aligned with the reference line except for the very extreme values, and therefore confirming that the threshold $k^* = 270$ and its associated Hill estimate seem to be well chosen.

Since the empirical quantile $\hat{q}_n(\beta_n) = X_{\lceil n\beta_n \rceil, n}$ underestimates the extreme quantile $q(\beta_n)$ for very extreme levels β_n as seen in the course, we rely on the Weissman (1978) estimator, which extrapolates from the intermediate threshold $X_{n-k,n}$ to the far tail using the Hill estimate $\hat{\gamma}_k$:

$$\hat{q}^*_n(\beta_n) := \left(\frac{n(1-\beta_n)}{k}\right)^{-\hat{\gamma}_k} X_{n-k,n} \tag{1}$$

An asymptotic 95% confidence interval is given by:

$$\hat{I}_n(\beta_n) = \left[\hat{q}^*_n(\beta_n) \times \exp\left(\pm \frac{\log\left(k/[n(1-\beta_n)]\right)}{\sqrt{k}} \times \hat{\gamma}_k \times 1.96\right)\right] \tag{2}$$

Using $k^* = 270$ and $\hat{\gamma}_{k^*} = 0.7306$, we estimate the extreme quantile at level $\beta_n = 1 - 1/n$. The results are presented in Table 3 and Figure 5. At the extreme level $\beta_n = 1 - 1/n \approx 0.9987$, the Weissman estimator predicts 6.02 million $[3.15, 8.89]$, and we have obtained for the empirical quantile $\beta_n = 1 - 1/n$ is 3.98 million. Hence, empirical quantile would severely underestimate the true extreme value city size. Note that the maximum sample size of 8.34 millions lies within the confidence interval of the Weissman estimate.

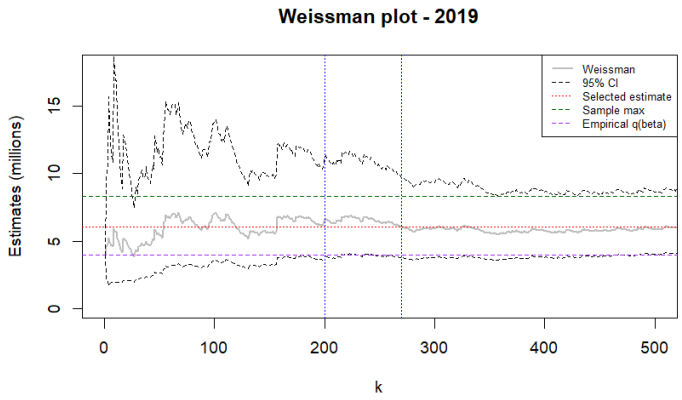


Figure 5: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2019 with $k^* = 270$.

Table 3: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2019 ($k = 270$, $\hat{\gamma} = 0.7136$).

Year	n	**Sample max (M)**	**Empirical** $q(\beta_n)$ **(M)**	**Weissman** $\hat{q}(\beta_n)$ **(M)**
2019	788	8.34	3.98	6.02

4.2 Replication Across All Years (2010–2019)

Having estimated the tail index of US city populations for 2019, we now replicate the same analysis across all years from 2010 to 2019 in order to study whether the extreme value structure of city population sizes has remained stable over the decade or not. Following the same methodology as for 2019, the Hill estimator is computed for each year from 2010 to 2019. For each year, the stable region is identified visually and the optimal k is selected manually. We then apply the Weissman estimator to obtain the estimated extreme quantiles and their empirical counterparts. All the results obtained are summarised in Table 4. First, we observe that across the three different estimations the tail index $\hat{\gamma}$ remains stable over time, with mean values of 0.7179 (Hill), 0.7085 (BR-Hill), and 0.7024 (dAMSE). This stability suggests that the heavy-tailed structure of US city populations is a persistent feature of the distribution. Regarding the extreme quantile estimates, the Weissman estimator at level $\beta_n = 1 - 1/n$ exceeds the empirical quantile at the same level across all years, ranging from 5.52 million (2013) to 6.58 million (2018), compared to empirical quantiles between 3.80 and 3.98 million. This confirms that here an empirical approach would severely underestimate the extreme city size. Note that the sample maximum, corresponding to New York City, ranging from 8.19 to 8.47 million, lies above the Weissman estimate. Taken together, all three estimators converge towards a tail index between approximately 0.70 and 0.73, confirming that the extreme behavior of US city population sizes is stable over time. Additionally, we have plotted the Exponential QQ-plot and the Pareto QQ-plot for each year (see Appendix, Section A.2), to check whether our selection of the stable region is appropriate. Across all years, both plots show that the points are approximately well aligned with the reference line, indicating that our selection of the stable region seems correct.

Table 4: Tail index and extreme quantile estimates at level $\beta_n = 1 - \frac{1}{n}$ by year (2010–2019)

Year	Stable region	k	Hill	BR-Hill	dAMSE	Max (M)	Empirical $q(\beta_n)$	Weissman $\hat{q}(\beta_n)$
2010	[190, 240]	240	0.7354	0.7321	0.7012	8.19	3.80	6.17
2011	[190, 240]	240	0.7390	0.7361	0.6966	8.27	3.82	6.33
2012	[260, 330]	330	0.7090	0.7018	0.6940	8.35	3.85	5.58
2013	[260, 350]	350	0.7047	0.6970	0.7065	8.40	3.88	5.52
2014	[270, 350]	350	0.7072	0.7003	0.7050	8.43	3.91	5.64
2015	[260, 380]	380	0.7074	0.6967	0.7011	8.46	3.94	5.70
2016	[220, 300]	300	0.7097	0.7004	0.7012	8.47	3.96	5.82
2017	[220, 260]	260	0.7189	0.7090	0.7026	8.44	3.98	6.11
2018	[160, 240]	240	0.7343	0.7198	0.7090	8.39	3.98	6.58
2019	[200, 270]	270	0.7136	0.6918	0.7064	8.34	3.98	6.02
Mean			**0.7179**	**0.7085**	**0.7024**			
SD			**0.0133**	**0.0155**	**0.0046**			

5 Extreme Value Analysis: Spatial Heterogeneity

To investigate whether the extreme value behavior of US city populations depends on geographic characteristics, we focus on the 2019 cross-section ($n = 788$ cities). Each city is treated as an independent observation. We map each city into one of the four regions defined by the official US Census Bureau regional classification[1]: Northeast ($n = 81$), West ($n = 290$), Midwest ($n = 162$), and South ($n = 247$).

5.1 Methodology

For each region, we compute the Hill and Bias-Reduced Hill estimators, select the stable region visually, and take k^* as the left part of the stable region in order to keep the ratio k^*/n intermediate and minimise bias. Given the sub-sample sizes, k^* is necessarily small. The selected values are reported in Table 5.

The validity of each threshold is then checked with an Exponential QQ-plot and a Pareto QQ-plot. Finally, we compute the Weissman extreme quantile estimator using the Bias-Reduced Hill estimate of γ.

5.2 Results

5.2.1 Hill plots and tail indexes

Table 5 reports the Hill and BR-Hill tail index estimates for each region. While $\hat{\gamma}$ remains positive across all four regions, confirming that heavy-tailed behaviour in US city populations is not specific to any geographic area, we observe heterogeneiety across the regions estimate compared to the full sample baseline ($\hat{\gamma} \approx 0.74$). In particular, the West lies below the baseline, the Midwest and Northeast above it. Strikingly, the South exhibits a substantial divergence between the raw Hill (0.715) and the BR-Hill (0.544). Since several regional subsamples are small, we take k^* as the lower bound of the stable region to ensure intermediate quantiles. However, we note that for Northeast ($k^*/n = 21\%$) and Midwest ($k^*/n = 18.5\%$), k is relatively large compared to the sample size.

[1] https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf

Table 5: Tail index estimates by US Census region (2019 cross-section)

Region	n	**Stable region**	k^*	k^*/n	**Hill**	**BR-Hill**
Northeast	81	[17, 23]	17	21%	0.7922	0.8021
West	290	[20, 40]	20	6.9%	0.6760	0.6735
Midwest	162	[30, 50]	30	18.5%	0.8042	0.8042
South	247	[23, 38]	23	9.3%	0.7150	0.5438
Full sample	780	[57, 150]	104	13.3%	0.7416	0.7332

Figure 6 displays the Hill and BR-Hill estimators as a function of k for each region. The blue vertical lines delimit the visually selected stable region, the red dotted horizontal line the regional BR-Hill estimate at k^*, and the navy horizontal line the full-sample BR-Hill baseline ($\hat{\gamma} \approx 0.74$). The West ($n = 290$) yields the most reliable estimate. Both curves stabilise quickly around $\hat{\gamma} \approx 0.67$ over $k \in [20, 40]$, with tight confidence bands and well-defined stable region. This is slightly below the baseline. The Midwest ($n = 162$) gives stabilise near $\hat{\gamma} \approx 0.80$ over $k \in [30, 50]$, slightly above the baseline. The South ($n = 247$) presents the most striking result. Indeed, while the raw Hill stabilises near the baseline around 0.715, the BR-Hill drops substantially and persistently to 0.544. This suggests significant upward bias in the raw Hill estimator, which likely driven by the internal heterogeneity of the region. Finally, the Northeast ($n = 81$) is the most problematic. The BR-Hill exceeds 1.5 for $k < 10$, driven by outsiders such as New York City (8.3 M), which is approximately 5 times larger than the next Northeastern city (Philadelphia, 1.6 M). The estimator only begins to stabilise around $k \in [17, 23]$, but at that point k is already large relative to the sample size (21%), so the resulting $\hat{\gamma} \approx 0.80$ must be interpreted with caution.

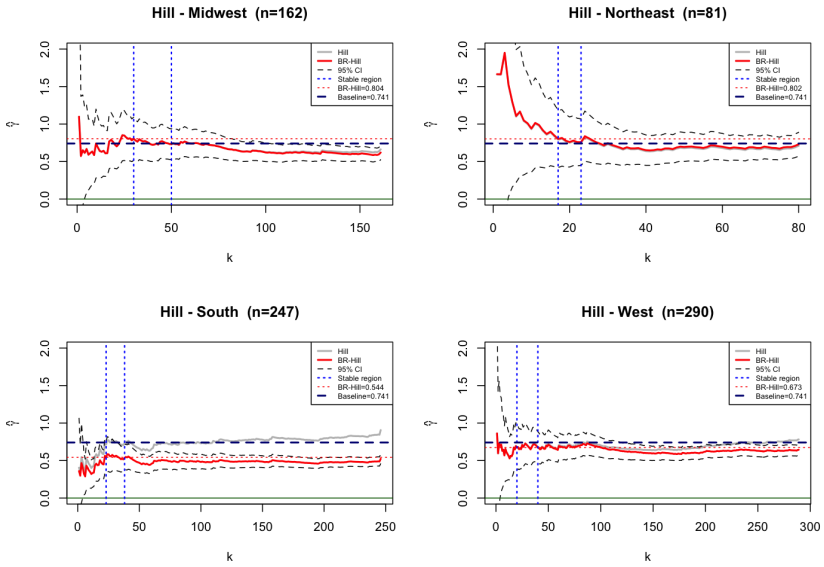


Figure 6: Hill and BR-Hill plots for the four US Census regions (2019). Blue dashed lines delimit the selected stable region; the red dotted line marks the regional BR-Hill at k^*; the navy line shows the full-sample BR-Hill baseline ($\hat{\gamma} \approx 0.74$).

5.2.2 Diagnostic QQ-Plots

Figure 43 in Section Appendix A.3 displays the exponential and Pareto QQ-plots for each geographic region, evaluated at the lower bound of the stable region (k goes from 17 to 30 across regions). For the Pareto-type tail behaviour, the QQ-plots provide reasonable support except for the Northeast, for which the higher quantiles

deviate substantially. However, this is expected since k is not truly intermediate (21% of the sample), and the deviation is likely induced by strong heterogeneity driven by New York and Philadelphia. In the exponential QQ-plots, the points align well along the reference line for the West and Midwest, suggesting a clean power-law tail at the selected threshold. However, the South empirical quantiles deviate upward. This suggests that the largest observations may be larger than what the fitted $\hat{\gamma}$ implies at $k = 23$. Taken together, these diagnostics support the use of the Pareto model for the tail of county-level population distributions across all regions except for the Northeast.

5.2.3 Weissman Extreme Quantile Estimates by Region

Figure 7 shows the Weissman extreme quantile estimates $\hat{Q}(1 - 1/n)$ by region, using the BR-Hill estimator at k^*. There is a common observation across all four regions: the selected estimate falls below the regional sample maximum. We can likely explain it because of small subsamples and city outliers. As a result, the estimator does not extrapolate far enough beyond the threshold to surpass the largest observed city at this level of β. The West produces the most stable and well-behaved curve, with a selected estimate around 2.3 M against a sample maximum of 4.0 M (Los Angeles), consistent with the clean tail diagnostics we observed earlier. The Midwest yields an estimate of approximately 2.0 M, below the sample maximum of 2.7 M (Chicago), with confidence bands that widen substantially for small k before stabilising. In the South, the low BR-Hill of $\hat{\gamma} = 0.544$ limits the extrapolation power of the estimator, producing a selected estimate of approximately 1.8 M against a sample maximum of 2.3 M (Houston). Finally, the Northeast remains the most problematic region. New York City (8.34 M) makes the sample max so high, and combined with a high k, that the selected estimate is only 1.3 M, substantially below. We observe the sensitivity of these estimates to more aggressive extrapolation levels in the Appendix (A.3).

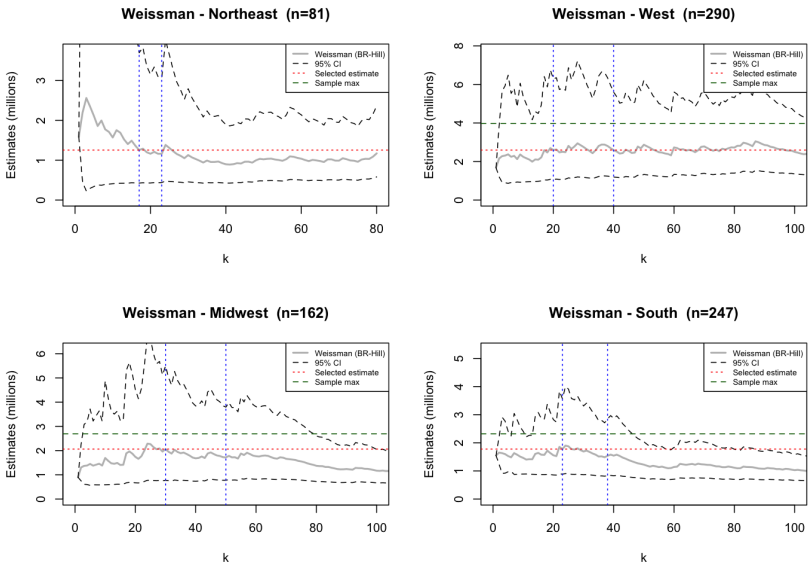


Figure 7: Weissman extreme quantile estimates $\hat{Q}(1 - 1/n)$ by US Census region (2019). Grey line: Weissman estimator using BR-Hill; dashed black lines: 95% confidence interval; red dotted line: selected estimate at k^*; green dashed line: sample maximum.

6 Predicting City Size in 2040

In this section, we will be interested in answering the following question: **Can we predict the size of a large US city in 2040?** To predict the size of a large city in the US in 2040, we will first for each of the US city in our sample, we fit a log-linear regression of the following form:

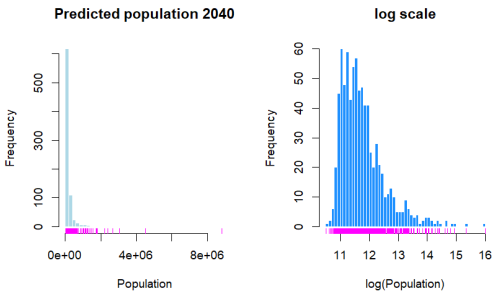
$$\log(\text{Pop}_{i,t}) = \beta_0^{(i)} + \beta_1^{(i)} \cdot t + \varepsilon_{i,t}$$

- $\text{Pop}_{i,t}$ is the population of city i in year t, with $t \in \{2010, 2011, \ldots, 2019\}$
- $\beta_0^{(i)}$ is the city-specific intercept.
- $\beta_1^{(i)}$ is the city-specific slope, interpreted as the average annual log-growth rate of city i.

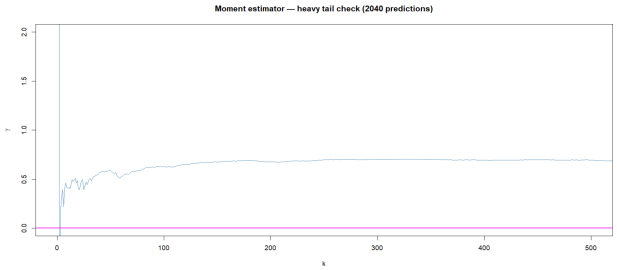
Each model is estimated by OLS. The out-of-sample prediction for 2040 is then obtained by the following extrapolation:

$$\widehat{\text{Pop}}_{i,2040} = \exp\left(\hat{\beta}_0^{(i)} + \hat{\beta}_1^{(i)} \times 2040\right)$$

Then after obtaining the predicted level of the population for 2040 for each city in our sample, we will check if the new distribution of the predicted population in 2040 is heavy tailed. According to the Figure 8a, we can observe that the distribution for the predicted level of population for 788 cities remains right-skewed, with a few very large cities driving a long upper tail. In addition, according to Figure 8b, the moment estimator $\hat{\gamma}$ plotted against k stabilizes and remains clearly above zero for a wide range of k values (up to k = 500), confirming that the 2040 predicted distribution is indeed **heavy-tailed**.



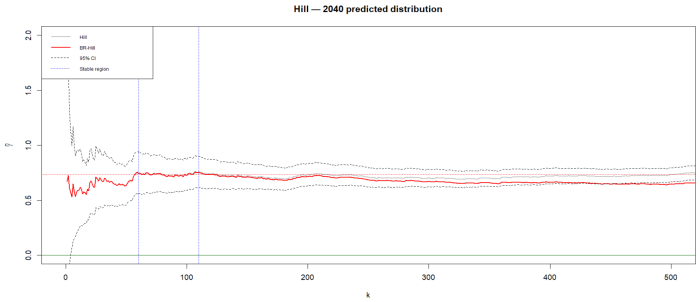
(a) Distribution of predicted city populations (raw and log-scale)



(b) Moment estimator of the Extreme Value Index

Figure 8: Heavy-tail confirmation for predicted US city populations in 2040.

Having demonstrated that the predicted population of US cities in 2040 follows a heavy-tailed distribution, we can estimate the Tail Index using the Hill estimator. We can therefore estimate the tail index using the Hill estimator and selecting manually the stable region. The tail index estimated on the predicted 2040 population distribution is very similar with the estimates obtained for the observed years 2010–2019. As reported in Table 9 (right) and Figure 9 (left), the Hill, BR-Hill and dAMSE estimators yield $\hat{\gamma} = 0.7370$, 0.7334 and 0.7086 respectively, all falling within the range observed over the historical period. This suggests that the heavy-tailed characteristic of US city population sizes is not only stable over the observed decade but is also expected to persist into 2040. Additionally, the choice of $k \in [60, 110]$ is validated by the exponential and Pareto QQ-plots displayed in Figure 45 in Section Appendix A.4, which show a satisfactory alignment with the reference line.



Stable k	Hill	BR-Hill	dAMSE
$[60, 110]$	0.7370	0.7334	0.7086

Figure 9: Hill plot and tail index estimates for the predicted US city population distribution in 2040.

Table 6 reports the Weissman extreme quantile estimate and the empirical quantile at the same level.

At the extreme level $\beta_n = 1 - 1/n \approx 0.9987$, the Weissman estimator yields a predicted maximum city size of 9.16 million inhabitants, compared to the empirical quantile of 4.52 million at the same level, this result could also been seen on the Figure 46 in the Appendix Section A.4. The Weissman estimate thus exceeds the empirical quantile by approximately 103%. Moreover, the Weissman estimate of 9.16 million slightly exceeds also the sample maximum of 8.83 million, which corresponds to the predicted population of New York City in 2040 under the log-linear trend assumption.

Table 6: Extreme quantile comparison for 2040 predictions (Weissman vs empirical)

Year	n	**Sample max (M)**	**Empirical** $q(1-1/n)$ **(M)**	**Weissman** $Q(1-1/n)$ **(M)**
2040	788	8.83	4.52	9.16

7 Conclusion

The analysis has studied the extreme value behavior of US city population size using annual frequency data for 788 cities over the period 2010-2019 and has addessed the following questions : the temporal stability of the tail index, the spatial heteogeneity of the extreme behavior across the US regions and the prediction of city sizes for 2040. Firstly, the data explanatory section has confirmed that the US city populations has a right-skewed distribution, following this 'many small/few huge pattern', providing evidence of a **heavy-tailed distribution**. Regarding temporal stability, the tail index $\hat{\gamma}$ estimated independently for each year from 2010 to 2019 remains very stable, suggesting that the heavy tailed is a persistent feature of the distribution. The spatial analysis confirms that all regions in the US are subject to heavy-tailed behavior. However, the tail index estimates vary substantially across the regions, around the full-sample baseline($\hat{\gamma} \approx 0.74$), which suggests heterogeneity. The West appears the most stable, while the Midwest and Northeast exhibit higher estimates. The South shows notable discrepancies between Hill and BR-Hill estimators, suggesting internal heterogeneity. The Northeast is the most problematic due to the dominance of New York City, which distorts the Pareto approximation and limits Weissman extrapolation. Overall, while the Pareto model seems appropriate, regional heterogeneity and outliers affect the reliability of extreme value estimates. Regarding the prediction of city sizes in 2040, the tail index estimated is 0.7370-0.7334 (Hill, BR-Hill), which is falling within the range observed over the historical period, suggesting that the heavy-tailed structure is expected to persist into 2040. However, this prediction rests on a strong assumption that each city growth rate observed between 2010 and 2019 remains constant over the next 20 years. In short, the results of this analysis point out that the US city population sizes follow a heavy tailed distribution. Extreme Value Theory prove here to be the appropriate framework to capture the extreme behavior, consistently outperforming the empirical quantile in the estimation of the extreme quantiles.

Appendix

A Additional Figures and Checks

A.1 Expolratory Data Analysis

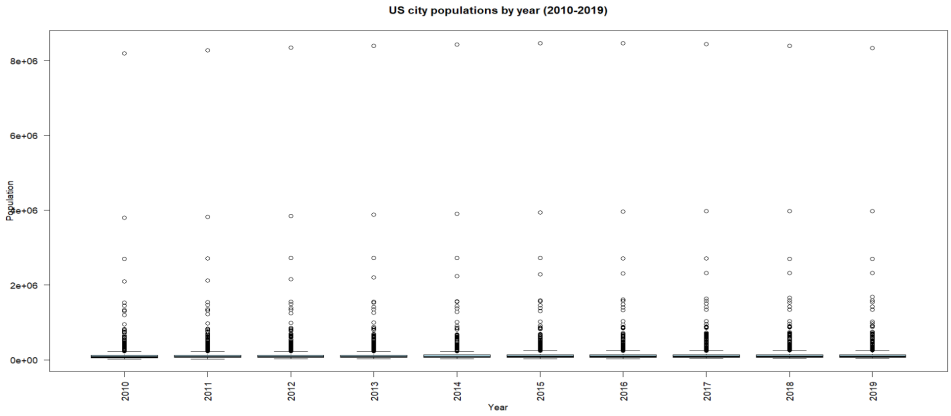


Figure 10: Boxplots for all years (2010–2019)

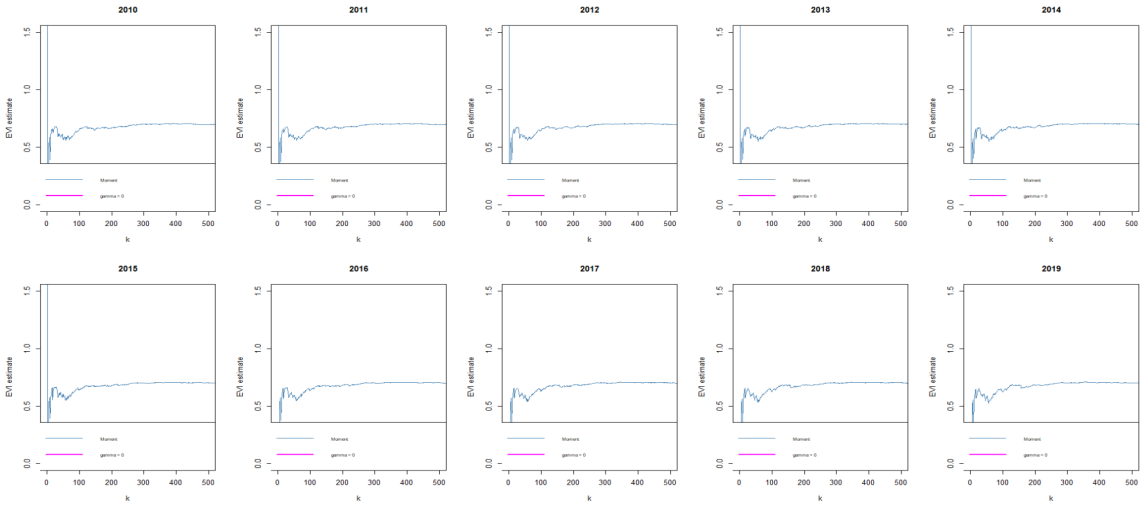


Figure 11: Moment estimator of the Extreme Value Index plotted against k, the number of top order statistics for all years (2010–2019)

A.2 Extreme Value Analysis : Temporal Stability of the Tail Index (2010-2019)

Hill Plots

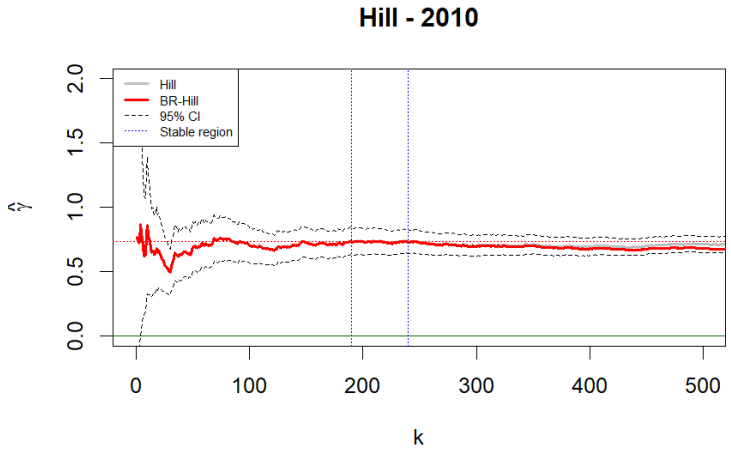


Figure 12: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2010.

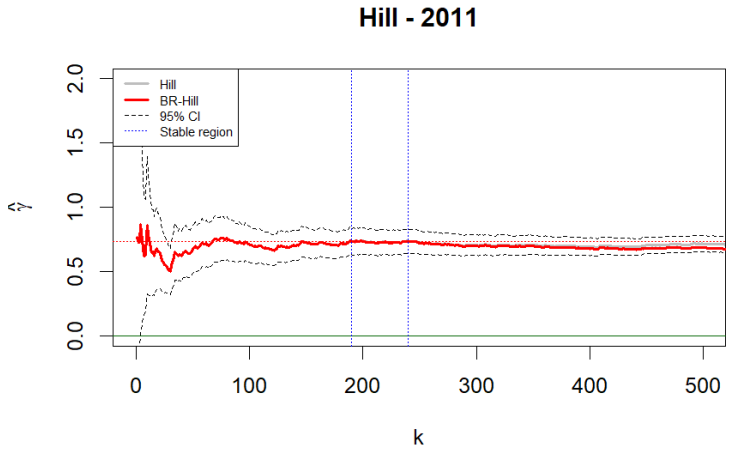


Figure 13: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2011.

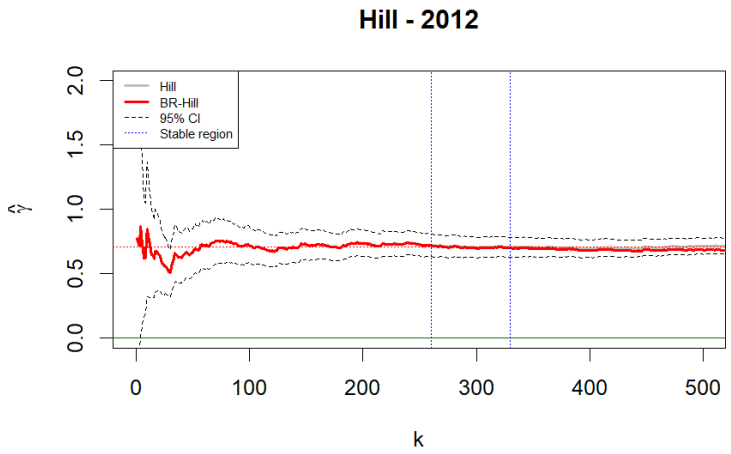


Figure 14: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2012.

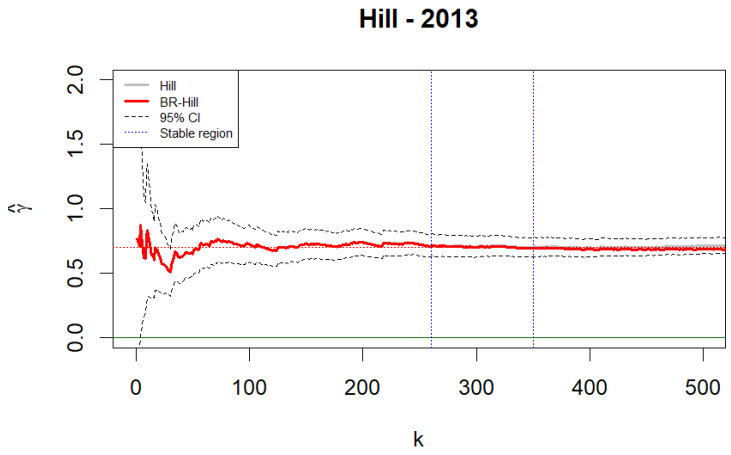


Figure 15: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2013.

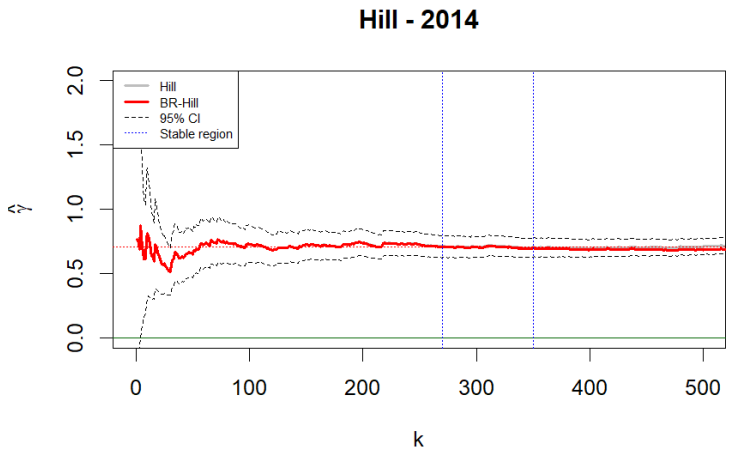


Figure 16: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2014.

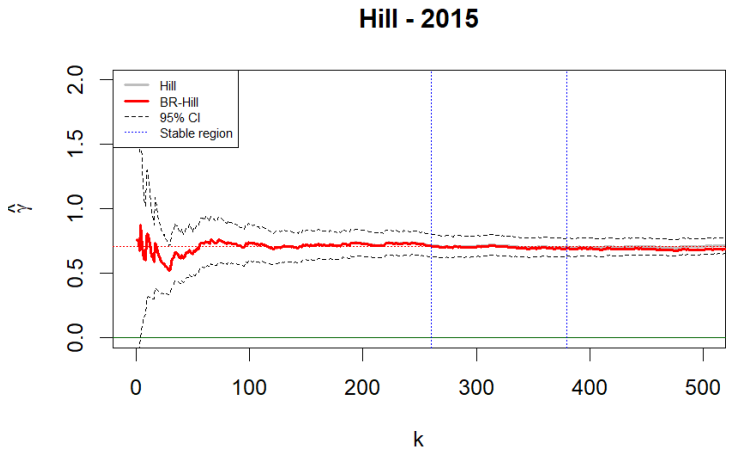


Figure 17: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2015.

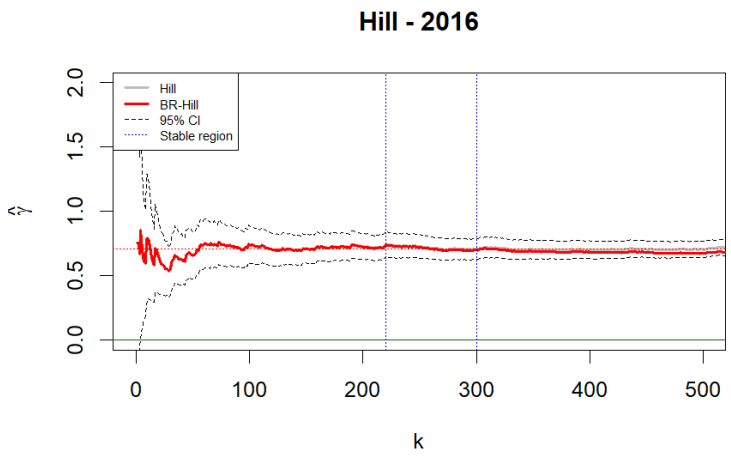


Figure 18: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2016.

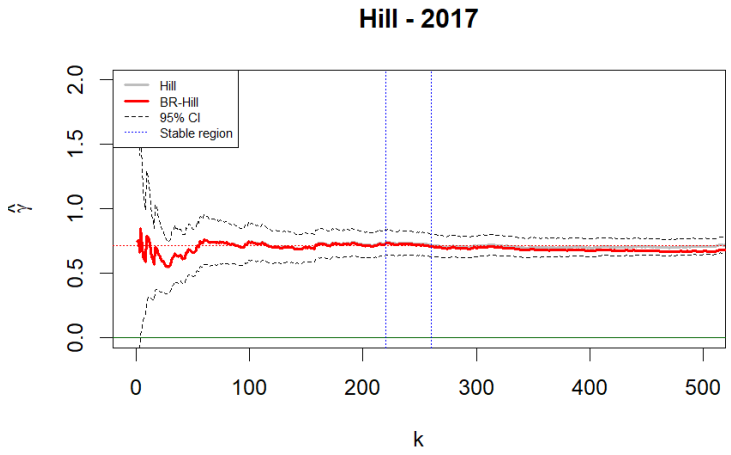


Figure 19: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2017.

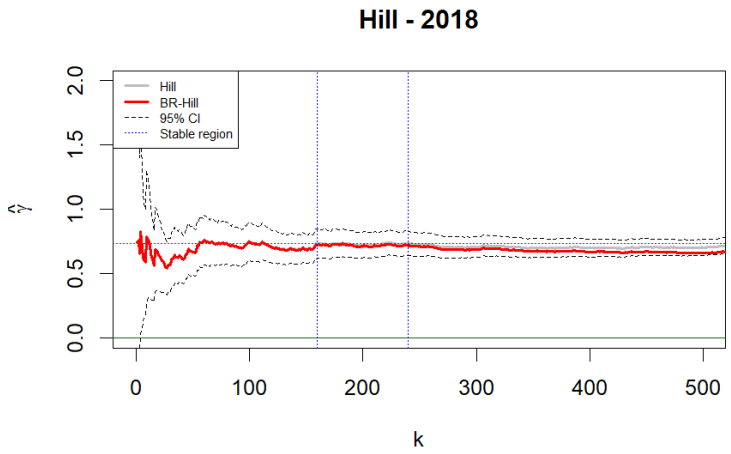


Figure 20: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the numer of top order statistics, for US city populations in 2018.

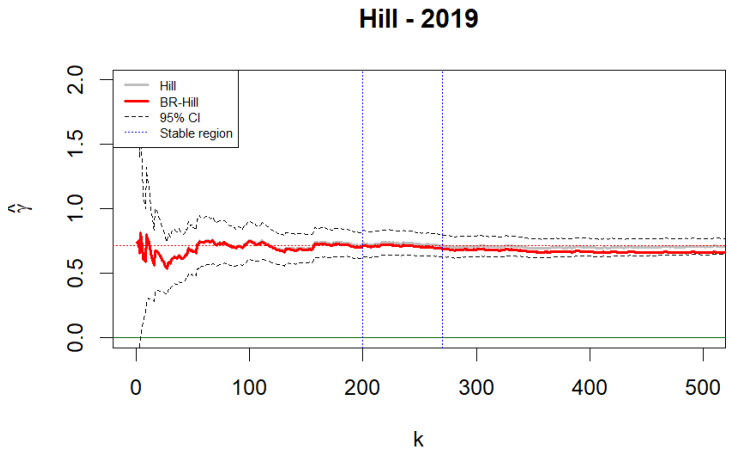


Figure 21: Hill plot of the Extreme Value Index $\hat{\gamma}$ against k, the number of top order statistics, for US city populations in 2019.

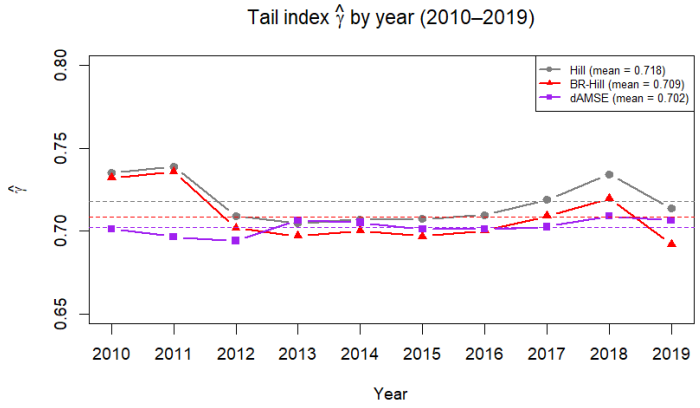


Figure 22: Tail index estimates $\hat{\gamma}$ by year (2010–2019) using the Hill (grey), BR-Hill (red) and dAMSE (purple) estimators.

Diagnostic Plots

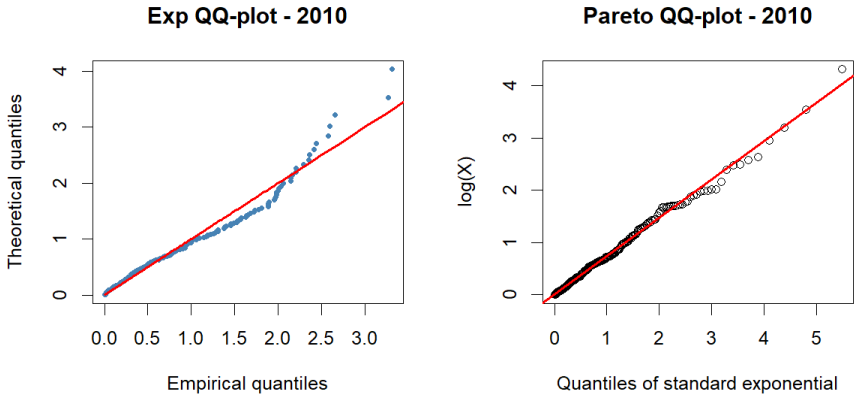


Figure 23: Exponential and Pareto Q–Q Plots for 2010.

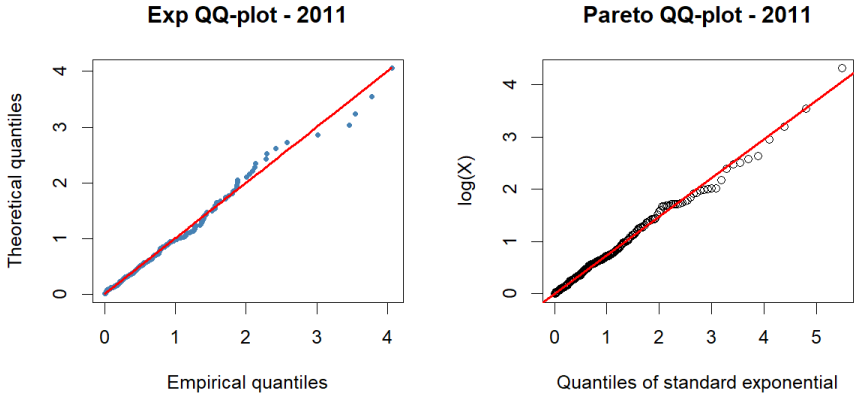


Figure 24: Exponential and Pareto Q–Q Plots for 2011.

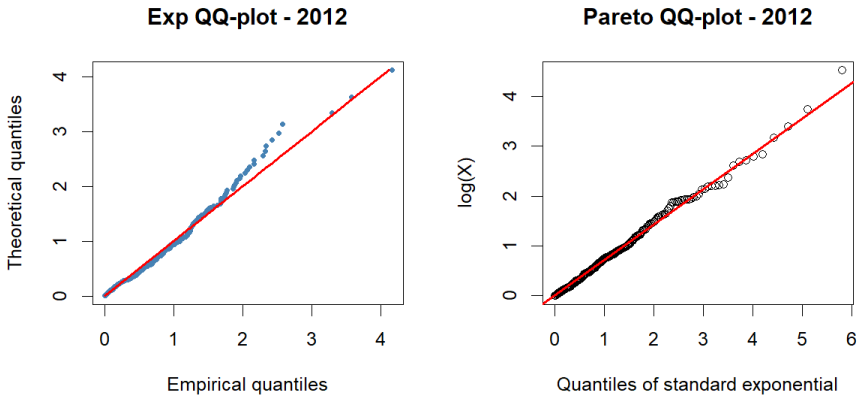


Figure 25: Exponential and Pareto Q–Q Plots for 2012.

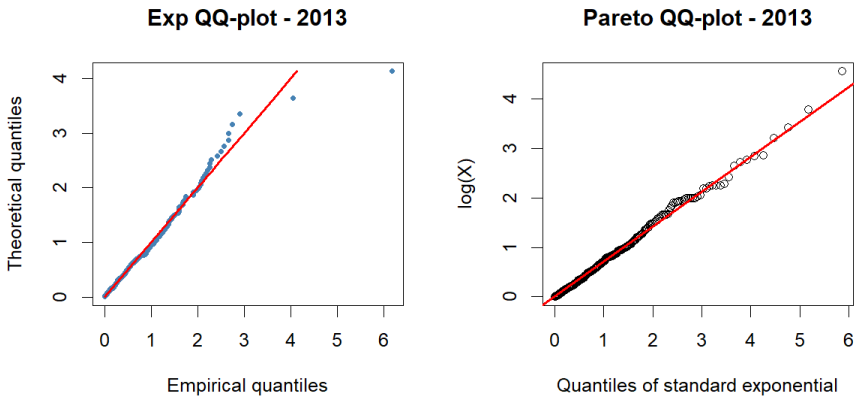


Figure 26: Exponential and Pareto Q–Q Plots for 2013.

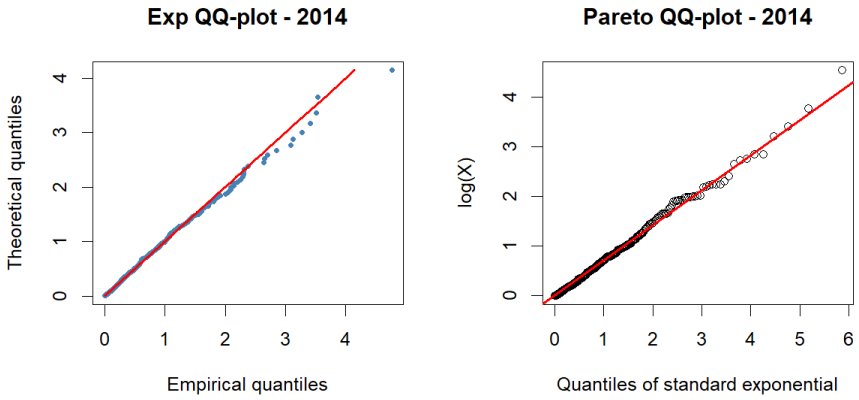


Figure 27: Exponential and Pareto Q–Q Plots for 2014.

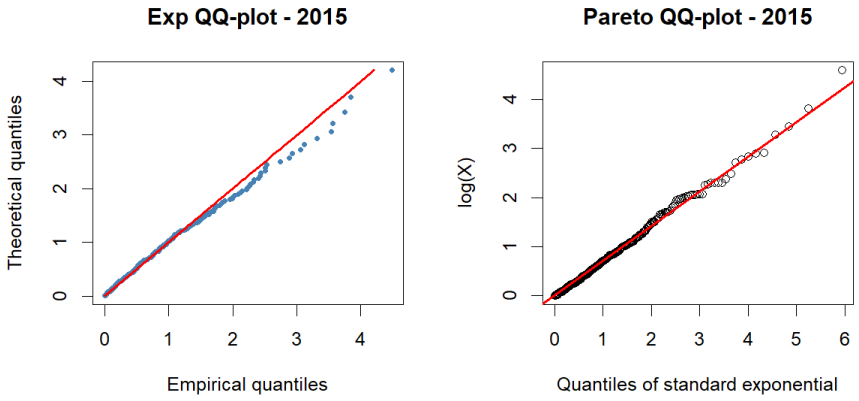


Figure 28: Exponential and Pareto Q–Q Plots for 2015.

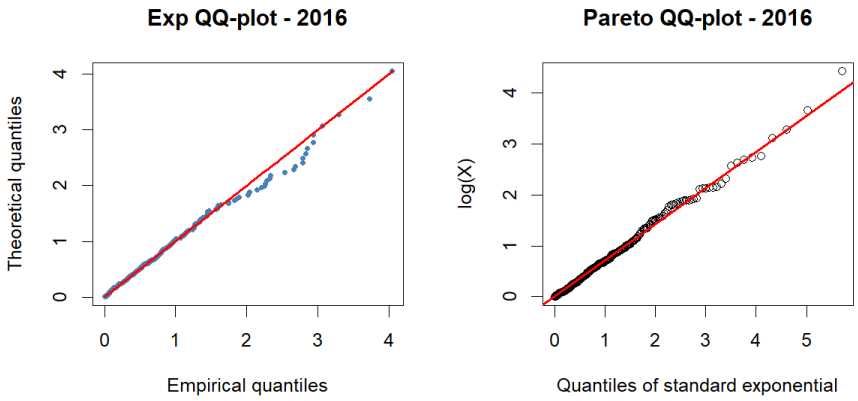


Figure 29: Exponential and Pareto Q–Q Plots for 2016.

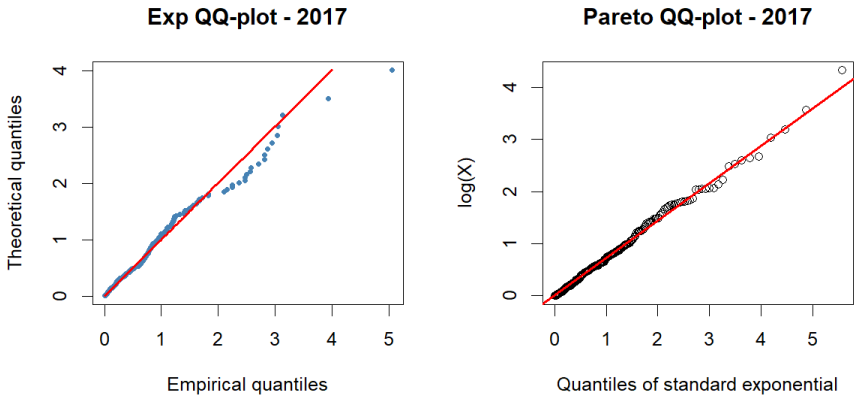


Figure 30: Exponential and Pareto Q–Q Plots for 2017.

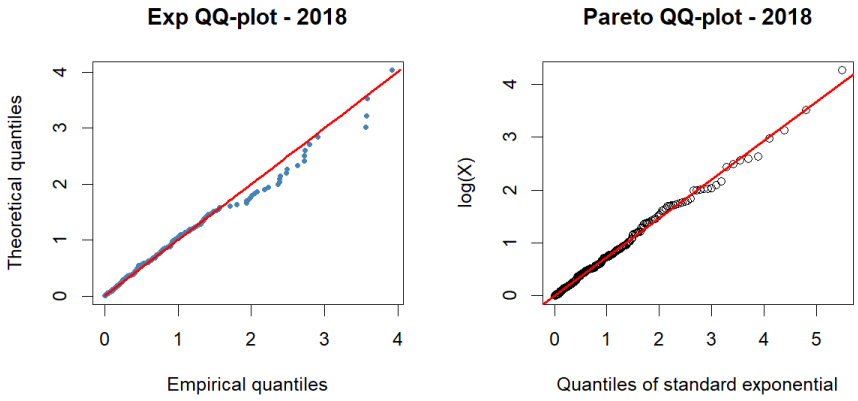


Figure 31: Exponential and Pareto Q–Q Plots for 2018.

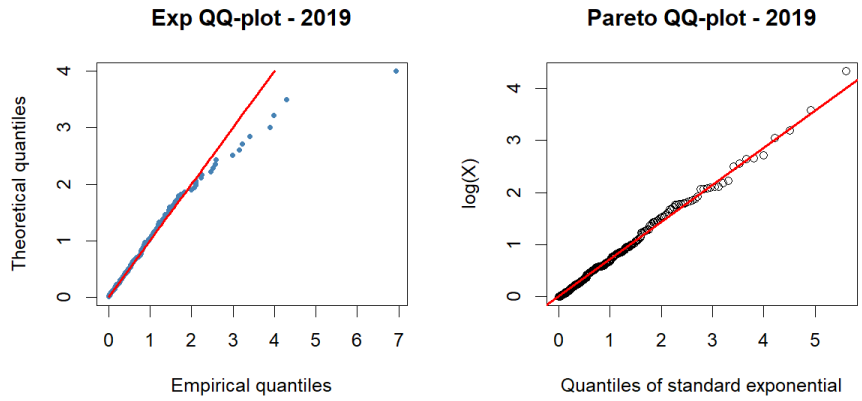


Figure 32: Exponential and Pareto Q–Q Plots for 2019.

Weissman extreme value quantile estimation

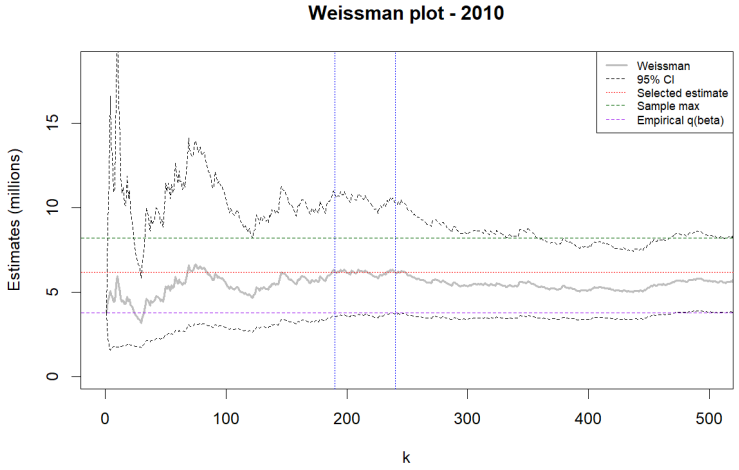


Figure 33: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2010.

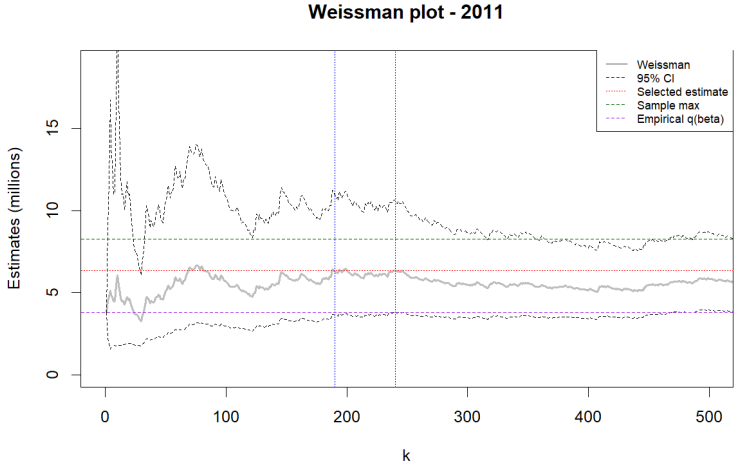


Figure 34: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2011.

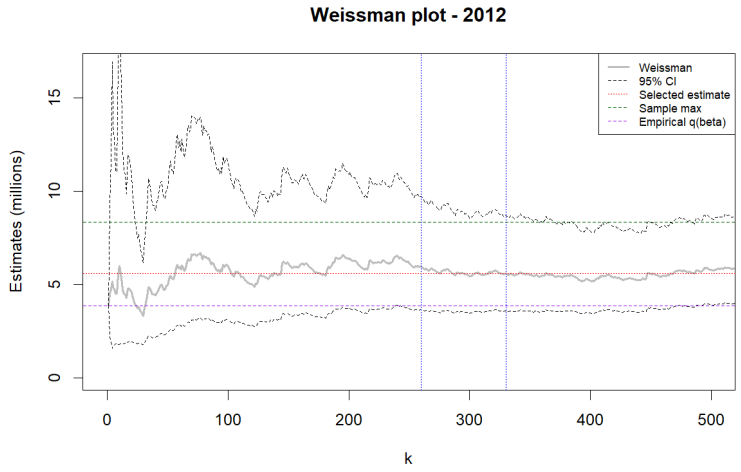


Figure 35: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2012.

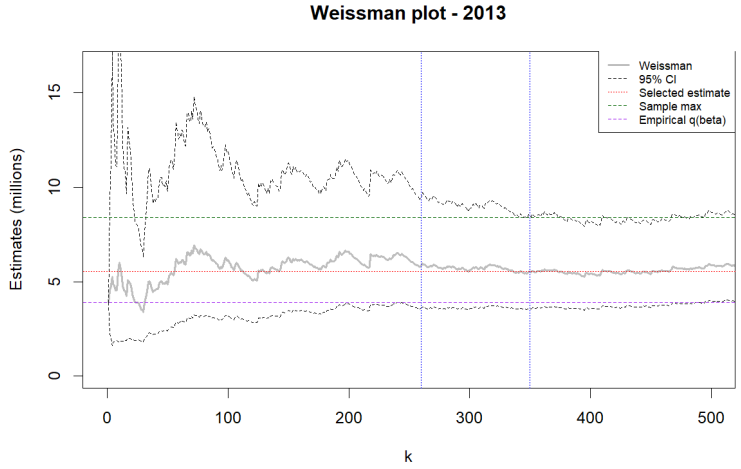


Figure 36: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2013.

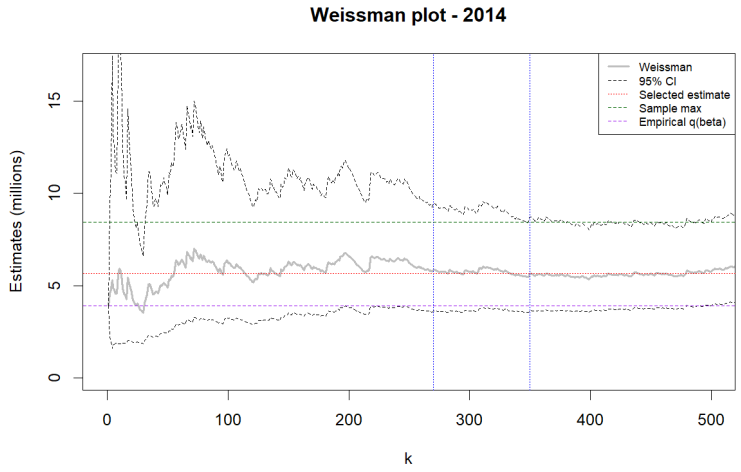


Figure 37: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2014.

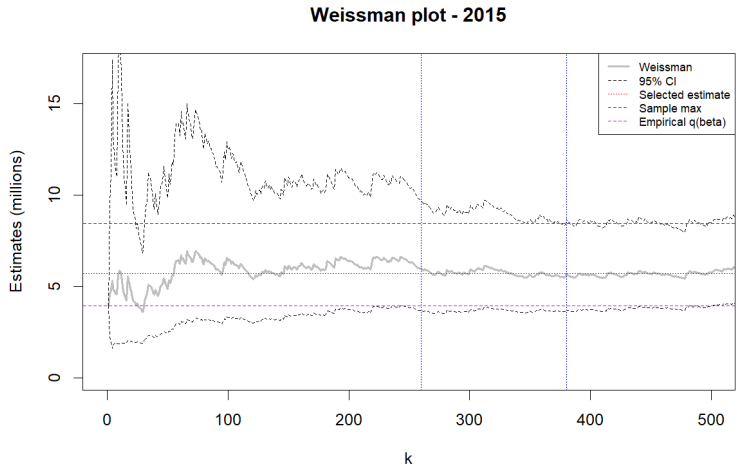


Figure 38: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2015.

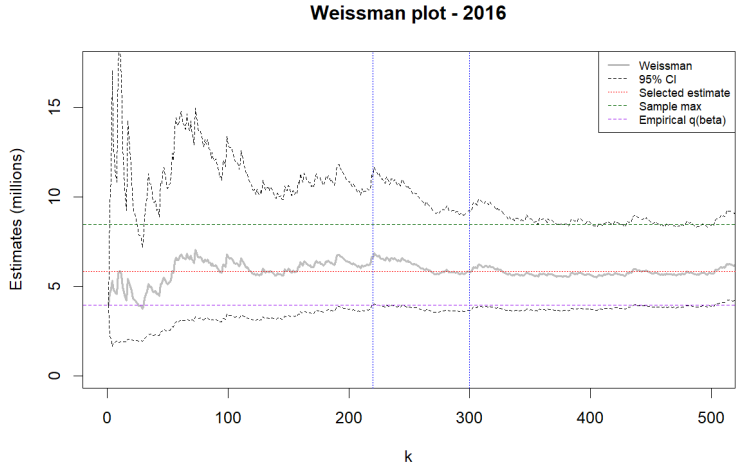


Figure 39: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2016.

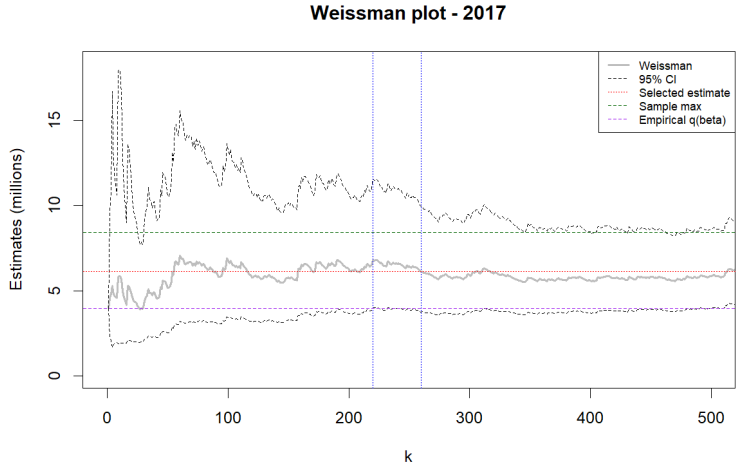


Figure 40: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2017.

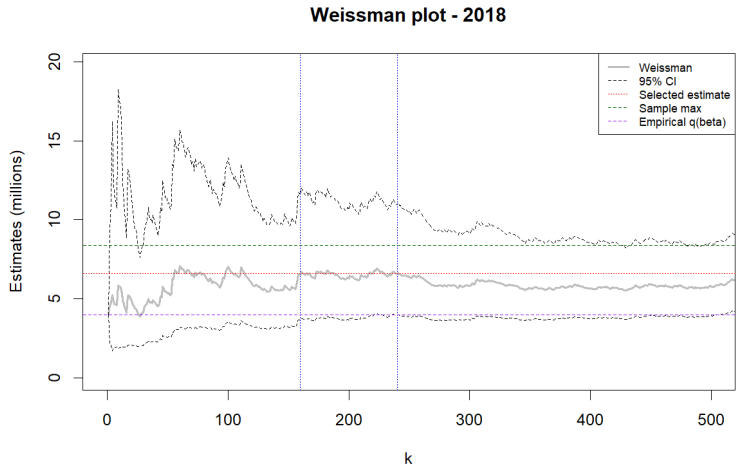


Figure 41: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2018.

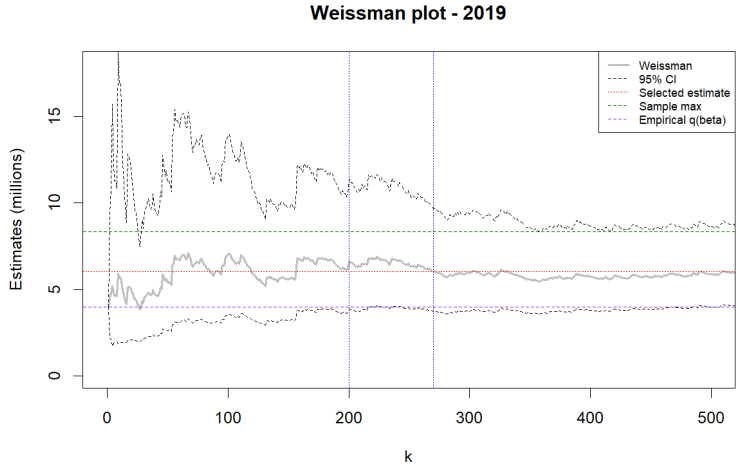
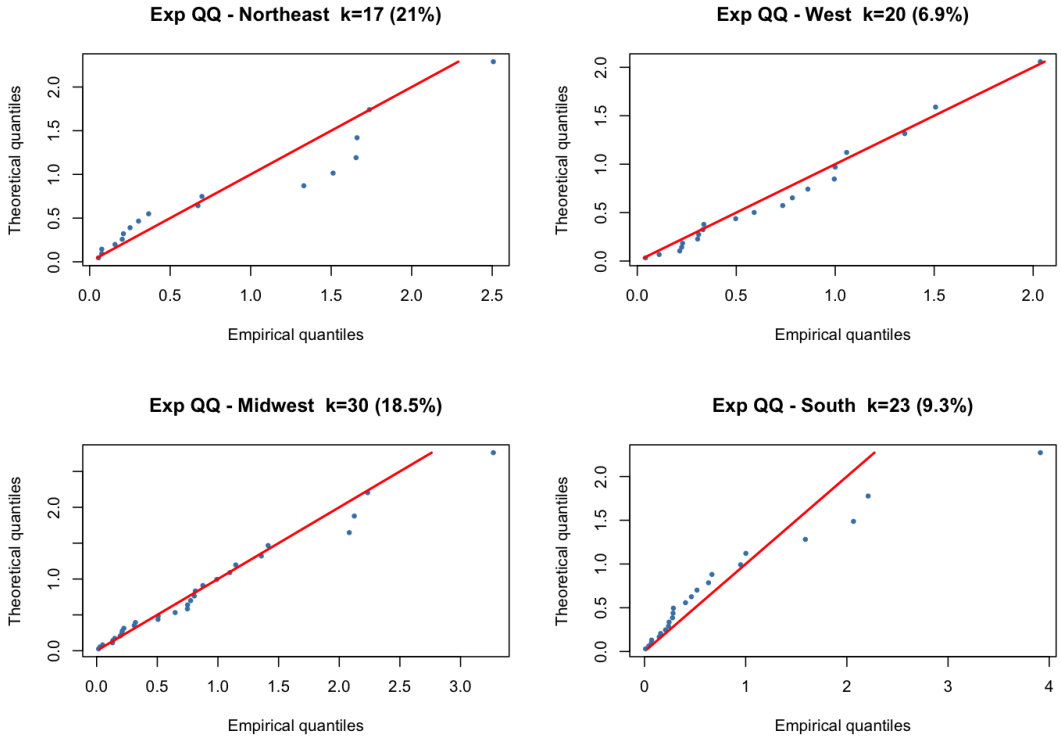


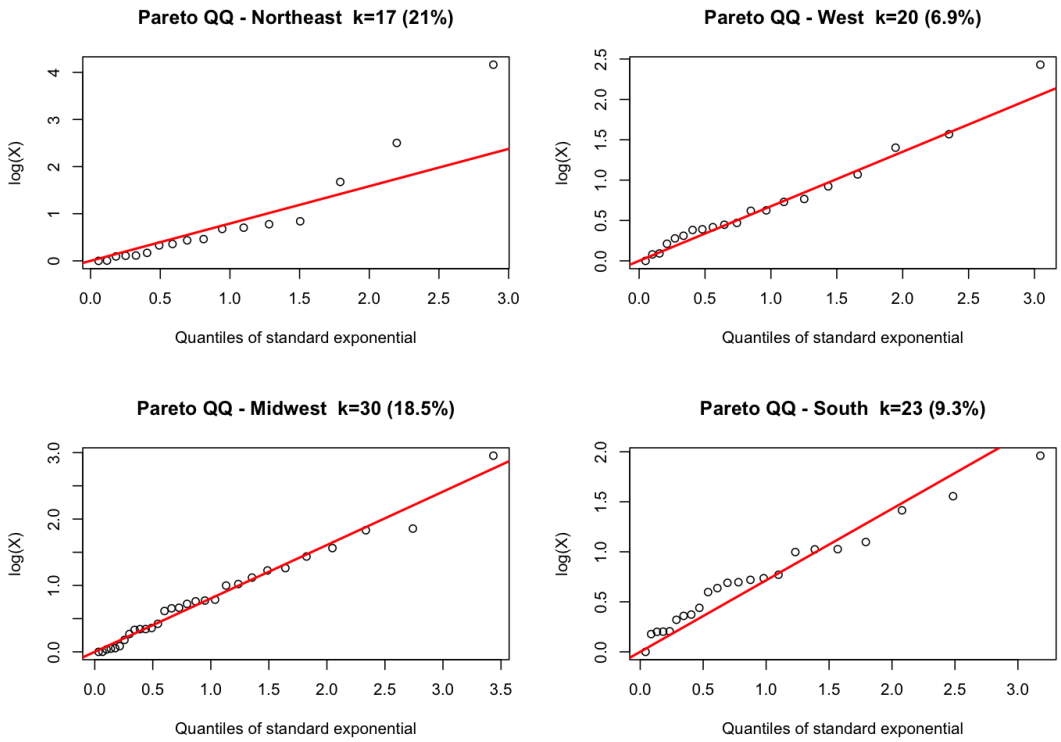
Figure 42: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2019.

A.3 Spatial Heterogeneity

Diagnostic Plots



(a) Exponential QQ-plots



(b) Pareto QQ-plots

Figure 43: Exponential QQ-plots (top) and Pareto QQ-plots (bottom) for the four US Census regions (2019). The selected k^* and the corresponding k^*/n ratio are indicated in each panel title.

Sensitivity to the extrapolation level

Figure 44 shows the Weissman estimator curves for four extrapolation levels $\beta \in \{1-1/n,\ 0.99,\ 0.995,\ 0.999\}$ across all regions. The sample maximum is shown as a black dashed line. First, the standard level $\beta = 1 - 1/n$ (grey) systematically falls below the sample maximum in all four regions, confirming the limitation discussed earlier: at this conservative extrapolation level, the estimator does not extrapolate far enough beyond the observed threshold to surpass the largest observed city. Second, only $\beta = 0.999$ (blue) manages to exceed the sample maximum, and only in the Midwest and Northeast — in the latter case however the curve is highly volatile and unstable around k^*, so this crossing should not be given much credence. Third, the South stands out as the region where *no* extrapolation level reaches the sample maximum of approximately 2.3 M (Houston): this is a direct consequence of the very low BR-Hill estimate of $\hat{\gamma} = 0.544$, which severely attenuates the extrapolation power of the Weissman estimator regardless of β. For the West, $\beta = 0.999$ approaches but does not clearly exceed the sample maximum of approximately 4.0 M (Los Angeles), with the curve remaining stable and well-behaved across k. Taken together, these results highlight that the regional subsamples are too small and the dominant urban outliers too influential for the Weissman estimator to produce reliable extrapolations beyond the observed maxima at conventional β levels. The Northeast in particular should be treated with caution at any extrapolation level given the instability of its tail index estimate.

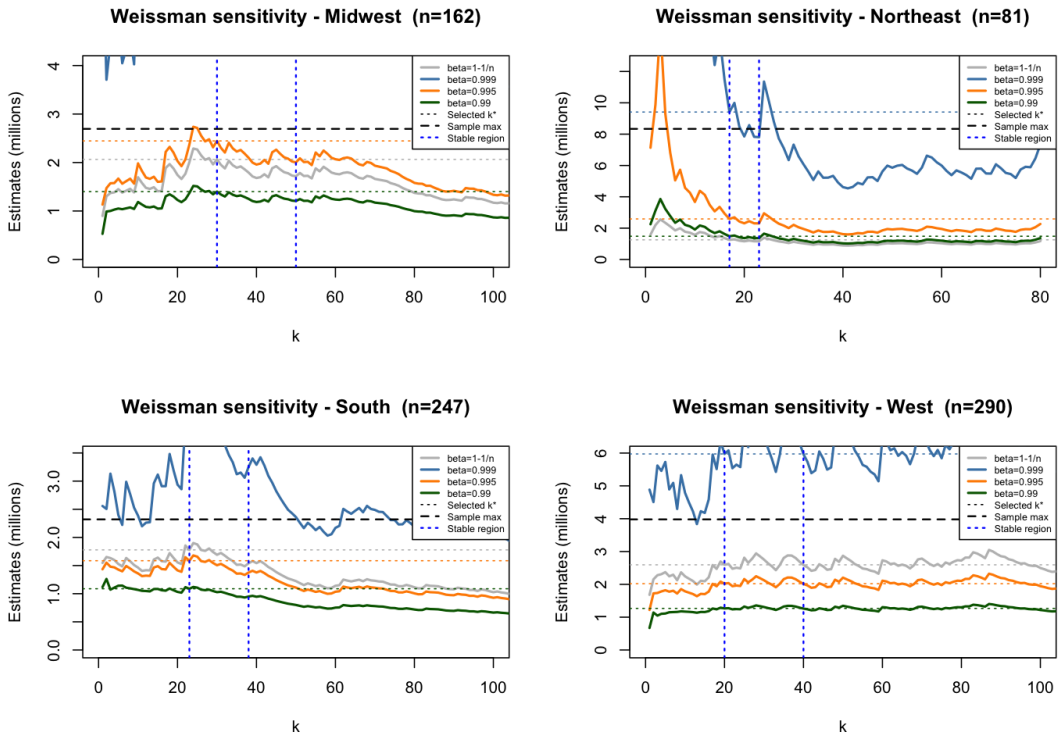


Figure 44: Weissman extreme quantile estimates for four extrapolation levels $\beta \in \{1 - 1/n,\ 0.99,\ 0.995,\ 0.999\}$ by US Census region (2019). Black dashed line: sample maximum. Blue dashed vertical lines: selected stable region $[lk, uk]$.

A.4 Predicting City Size in 2040

Table 7: Fastest growing and declining cities (log-linear trend)

Top 5 Growing Cities			Top 5 Declining Cities		
City	**Growth**	**2040**	**City**	**Growth**	**2040**
Leander city, Texas	0.09274	416,955	Charleston city, West Virginia	-0.01108	37,210
Herriman city, Utah	0.09248	331,703	Shreveport city, Louisiana	-0.00899	156,177
Little Elm city, Texas	0.08245	303,436	Jackson city, Mississippi	-0.00899	135,102
Queen Creek town, Arizona	0.07504	239,349	Decatur city, Illinois	-0.00893	58,598
Frisco city, Texas	0.06028	704,304	Albany city, Georgia	-0.00889	59,982

Diagnostic Plots

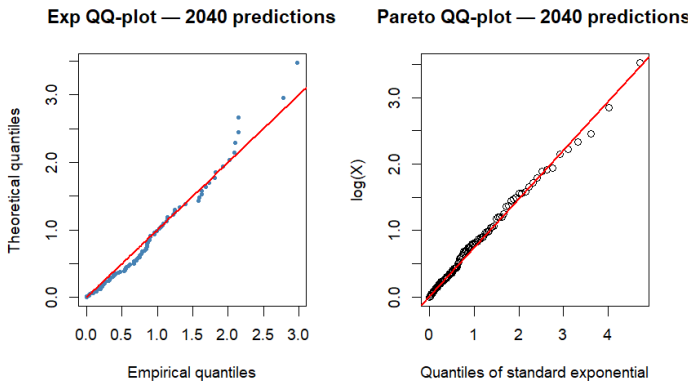


Figure 45: Exponential QQ plot and Pareto QQ plot with $k*$ = 110 for 2040 prediction.

A.5 Weissman extreme quantile estimation

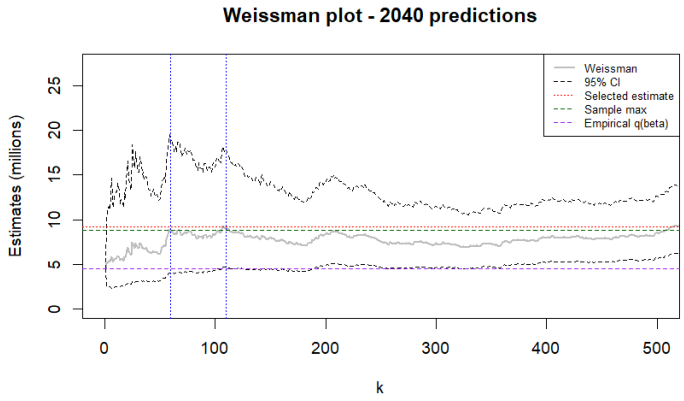


Figure 46: Weissman extreme quantile estimate at level $\beta_n = 1 - 1/n$ for year 2040 prediction.

B Relevant Code Excerpts

B.1 Data Preparation

```
raw <- readxl::read_excel("C:/Users/Victoria/Downloads/census2010.xlsx",
                          skip = 3, col_names = FALSE)
colnames(raw) <- c("Rank", "City", "Census2010", "EstBase",
                   paste0("Pop", 2010:2019))
df <- raw %>%
  mutate(
    Rank       = suppressWarnings(as.numeric(Rank)),
    Census2010 = suppressWarnings(as.numeric(Census2010))
  ) %>%
  filter(!is.na(Rank))
for (yr in 2010:2019) {
  col <- paste0("Pop", yr)
  df[[col]] <- suppressWarnings(as.numeric(df[[col]]))
}
cat("Cities loaded:", nrow(df))
```

B.2 Exploratory Data Analysis

Exploratory Data Analysis

```
# Overall pooled stats
x_pool <- unlist(lapply(2010:2019, function(yr) {
  x <- df[[paste0("Pop", yr)]]
  x[!is.na(x) & x > 0]
}))
overall_stats <- data.frame(
  Sample = "Pooled 2010-2019",
  n      = length(x_pool),
  Min    = round(min(x_pool),              0),
  Max    = round(max(x_pool),              0),
  Mean   = round(mean(x_pool),             0),
  Median = round(median(x_pool),           0),
  SD     = round(sd(x_pool),               0),
  P95    = round(quantile(x_pool, 0.95),   0),
  P99    = round(quantile(x_pool, 0.99),   0)
)
knitr::kable(overall_stats,
             col.names = c("Sample", "n", "Min", "Max", "Mean",
                           "Median", "SD", "P95", "P99"),
             caption   = "Overall pooled summary statistics (2010-2019)",
             align     = rep("c", 9),
             format.args = list(big.mark = ","))
# Annual stats
annual_stats <- data.frame()
for (yr in 2010:2019) {
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr <- x_yr[!is.na(x_yr) & x_yr > 0]
  annual_stats <- rbind(annual_stats, data.frame(
    Year   = yr,
    n      = length(x_yr),
    Min    = round(min(x_yr),           0),
    Max    = round(max(x_yr),           0),
    Mean   = round(mean(x_yr),          0),
    Median = round(median(x_yr),        0),
    SD     = round(sd(x_yr),            0),
    P95    = round(quantile(x_yr, 0.95), 0),
    P99    = round(quantile(x_yr, 0.99), 0)
  ))
}
knitr::kable(annual_stats,
             col.names = c("Year", "n", "Min", "Max", "Mean",
```

```
                              "Median", "SD", "P95", "P99"),
             caption   = "Annual summary statistics by year (populations)",
             align     = rep("c", 9),
             format.args = list(big.mark = ","))

years <- 2010:2019
par(mfrow = c(2, 5))
# Raw data histograms
for (yr in years) {
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr <- x_yr[!is.na(x_yr) & x_yr > 0]
  hist(x_yr, nclass = 50,
       main = as.character(yr),
       xlab = "Population",
       col  = "lightblue", freq = FALSE)
  rug(x_yr, col = "magenta")
}
par(mfrow = c(1, 1))
# Log scale histograms
par(mfrow = c(2, 5))
for (yr in years) {
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr <- x_yr[!is.na(x_yr) & x_yr > 0]
  hist(log(x_yr), nclass = 40,
       main = paste("log -", yr),
       xlab = "log(Population)",
       col  = "dodgerblue", freq = FALSE)
  rug(log(x_yr), col = "magenta")}
par(mfrow = c(1, 1))
# Boxplot
pop_list       <- lapply(years, function(yr) {
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr[!is.na(x_yr) & x_yr > 0]
})
names(pop_list) <- years
boxplot(pop_list,
        main  = "US city populations by year (2010-2019)",
        ylab  = "Population",
        xlab  = "Year",
        col   = "lightblue",
        names = years,
        las   = 2)
# Moment estimator
par(mfrow = c(2, 5))
for (yr in years) {
```

```
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr <- x_yr[!is.na(x_yr) & x_yr > 0]
  mom <- Moment(x_yr[x_yr > 0])
  plot(mom$gamma, type = "l", lwd = 1.5,
       ylim = c(0, 1.5), xlim = c(0, 500),
       xlab = "k", ylab = "EVI estimate",
       main = as.character(yr),
       col  = "steelblue")
  abline(h = 0, col = "magenta", lty = 1, lwd = 2)
  legend("bottomright", cex = 0.6,
         legend = c("Moment", "gamma = 0"),
         col    = c("steelblue", "magenta"),
         lty    = 1, lwd = c(1.5, 2))
}
par(mfrow = c(1, 1))

# Just 2019 to display
x2019 <- df$Pop2019
x2019 <- x2019[!is.na(x2019) & x2019 > 0]
n2019 <- length(x2019)
# Histogram
par(mfrow = c(1, 2))
hist(x2019, nclass = 50,
     main = "Population - 2019",
     xlab = "Population",
     col  = "lightblue", freq = FALSE)
rug(x2019, col = "magenta")
hist(log(x2019), nclass = 40,
     main = "log(Population) - 2019",
     xlab = "log(Population)",
     col  = "dodgerblue", freq = FALSE)
rug(log(x2019), col = "magenta")
par(mfrow = c(1, 1))
# Boxplot
boxplot(x2019,
        main = "Boxplot - US city populations (2019)",
        ylab = "Population",
        col  = "lightblue")
# Moment estimator : 2010, 2014, 2019
par(mfrow = c(1, 3))
for (yr in c(2010, 2014, 2019)) {
  x_yr <- df[[paste0("Pop", yr)]]
  x_yr <- x_yr[!is.na(x_yr) & x_yr > 0]
  mom <- Moment(x_yr[x_yr > 0])
  plot(mom$gamma, type = "l", lwd = 1.5,
```

```
       ylim = c(0, 1.5), xlim = c(0, 500),
       xlab = "k", ylab = "EVI estimate",
       main = paste("Moment estimator -", yr),
       col  = "steelblue")
  abline(h = 0, col = "magenta", lty = 1, lwd = 2)
  legend("bottomright", cex = 0.7,
         legend = c("Moment estimator", "gamma = 0"),
         col    = c("steelblue", "magenta"),
         lty    = 1, lwd = c(1.5, 2))
}
par(mfrow = c(1, 1))
```

B.3 Why Modelling the Bulk is not Approriate?

Fitting distributions

```
library(fitdistrplus)
library(actuar)
if ("ReIns"    %in% (.packages())) detach("package:ReIns",    unload = TRUE)
if ("tea"      %in% (.packages())) detach("package:tea",      unload = TRUE)
if ("extRemes" %in% (.packages())) detach("package:extRemes", unload = TRUE)
dt_ls <- function(x, df=1, mu=0, sigma=1) 1/sigma * dt((x-mu)/sigma, df)
pt_ls <- function(q, df=1, mu=0, sigma=1) pt((q-mu)/sigma, df)
qt_ls <- function(p, df=1, mu=0, sigma=1) qt(p, df)*sigma + mu
rt_ls <- function(n, df=1, mu=0, sigma=1) rt(n, df)*sigma + mu
save_dir <- "C:/Users/Victoria/Downloads/"
save_2panels <- function(fit, x, distname, filename) {
  png(paste0(save_dir, filename, ".png"),
      width = 1400, height = 700, res = 120)
  par(mfrow = c(1, 2))
  hist(x, nclass = 50, freq = FALSE,
       main = paste("Histogram and", distname, "density"),
       xlab = "data", col = "white", border = "grey40")
  xseq <- seq(min(x), quantile(x, 0.99), length.out = 500)
  dens <- switch(distname,
    Normal    = dnorm(xseq,
                      mean = fit$estimate["mean"],
                      sd   = fit$estimate["sd"]),
    Lognormal = dlnorm(xseq,
                       meanlog = fit$estimate["meanlog"],
                       sdlog   = fit$estimate["sdlog"]),
    Loggamma  = {
      lx <- log(x)
      dlnorm(xseq,
             meanlog = mean(lx),
             sdlog   = sd(lx))
    },
    Pareto    = dpareto(xseq,
                        shape = fit$estimate["shape"],
                        scale = fit$estimate["scale"]),
    Cauchy    = dcauchy(xseq,
                        location = fit$estimate["location"],
                        scale    = fit$estimate["scale"]),
    Student   = dt_ls(xseq,
                      df    = fit$estimate["df"],
                      mu    = fit$estimate["mu"],
                      sigma = fit$estimate["sigma"]),
```

```
  Burr      = dburr(xseq,
                    shape1 = fit$estimate["shape1"],
                    shape2 = fit$estimate["shape2"],
                    scale  = fit$estimate["scale"]),
  Frechet   = dfrechet(xseq,
                       shape = fit$estimate["shape"],
                       scale = fit$estimate["scale"]),
  NULL
)
if (!is.null(dens)) lines(xseq, dens, col = "red", lwd = 2)
n_x   <- length(x)
probs <- (1:n_x) / (n_x + 1)
emp_q <- sort(x)
theo_q <- switch(distname,
  Normal    = qnorm(probs,
                    mean = fit$estimate["mean"],
                    sd   = fit$estimate["sd"]),
  Lognormal = qlnorm(probs,
                     meanlog = fit$estimate["meanlog"],
                     sdlog   = fit$estimate["sdlog"]),
  Loggamma  = qlnorm(probs,
                     meanlog = mean(log(x)),
                     sdlog   = sd(log(x))),
  Pareto    = qpareto(probs,
                      shape = fit$estimate["shape"],
                      scale = fit$estimate["scale"]),
  Cauchy    = qcauchy(probs,
                      location = fit$estimate["location"],
                      scale    = fit$estimate["scale"]),
  Student   = qt_ls(probs,
                    df    = fit$estimate["df"],
                    mu    = fit$estimate["mu"],
                    sigma = fit$estimate["sigma"]),
  Burr      = qburr(probs,
                    shape1 = fit$estimate["shape1"],
                    shape2 = fit$estimate["shape2"],
                    scale  = fit$estimate["scale"]),
  Frechet   = qfrechet(probs,
                       shape = fit$estimate["shape"],
                       scale = fit$estimate["scale"]),
  NULL
)
if (!is.null(theo_q)) {
  plot(theo_q, emp_q,
```

```
        main = "Q...Q plot",
        xlab = "Theoretical quantiles",
        ylab = "Empirical quantiles",
        pch  = 1, col = "red", cex = 0.6)
    abline(0, 1, col = "black", lwd = 2)
  }
  par(mfrow = c(1, 1))
  dev.off()
  cat("Saved:", filename, "\n")
}
for (yr in c(2010, 2014, 2019)) {
  cat("\n===== YEAR:", yr, "=====\n")
  x <- df[[paste0("Pop", yr)]]
  x <- x[!is.na(x) & x > 0]
  # Normal
  fit <- tryCatch(fitdist(x, "norm", method = "mle"),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Normal",
                                  paste0(yr, "_00_normal"))
  # Lognormal
  fit <- tryCatch(fitdist(x, "lnorm", method = "mle"),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Lognormal",
                                  paste0(yr, "_01_lognormal"))
  # Log-gamma
  lx  <- log(x)
  fit <- tryCatch(fitdist(lx, "gamma", method = "mle"),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Loggamma",
                                  paste0(yr, "_02_loggamma"))
  # Pareto
  fit <- tryCatch(fitdist(x, "pareto",
                          start  = list(shape = 1.5, scale = min(x) * 0.99),
                          method = "mle", lower = c(0.01, 0)),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Pareto",
                                  paste0(yr, "_03_pareto"))
  # Cauchy
  fit <- tryCatch(fitdist(x, "cauchy",
                          start  = list(location = median(x), scale = IQR(x)/2),
                          method = "mle"),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Cauchy",
                                  paste0(yr, "_04_cauchy"))
```

```
  # Student-t location-scale
  fit <- tryCatch(suppressWarnings(
                   fitdist(x, "t_ls",
                           start = list(df = 3, mu = mean(x), sigma = sd(x)))),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Student",
                                  paste0(yr, "_05_student"))
  # Burr
  fit <- tryCatch(fitdist(x, "burr",
                          start   = list(shape1 = 2, shape2 = 1,
                                         scale  = median(x) * 0.5),
                          method  = "mle", lower = c(0.01, 0.01, 0),
                          control = list(maxit = 5000, reltol = 1e-10)),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Burr",
                                  paste0(yr, "_06_burr"))
  # Frechet
  fit <- tryCatch(fitdist(x, "frechet",
                          start  = list(shape = 1/0.7, scale = median(x)),
                          method = "mle", lower = c(0.01, 0)),
                  error = function(e) NULL)
  if (!is.null(fit)) save_2panels(fit, x, "Frechet",
                                  paste0(yr, "_07_frechet"))
  cat("Done for year", yr, "\n")
}
library(ReIns)
library(tea)
library(extRemes)
```

Quantification for 2019 year

```
x <- df[["Pop2019"]]
x <- x[!is.na(x) & x > 0]
fit_norm  <- tryCatch(fitdist(x, "norm",    method = "mle"),
error = function(e) NULL)
fit_ln    <- tryCatch(fitdist(x, "lnorm",   method = "mle"),
error = function(e) NULL)
fit_lg    <- tryCatch(fitdist(log(x), "gamma", method = "mle"),
error = function(e) NULL)
fit_par   <- tryCatch(fitdist(x, "pareto",
                              start  = list(shape = 1.5, scale = min(x) * 0.99),
                              method = "mle", lower = c(0.01, 0)),
                              error = function(e) NULL)
fit_cau   <- tryCatch(fitdist(x, "cauchy",
                              start  = list(location = median(x),
```

```
                              scale = IQR(x)/2),
                              method = "mle"), error = function(e) NULL)
fit_stu   <- tryCatch(suppressWarnings(
                        fitdist(x, "t_ls",
                                start = list(df = 3, mu = mean(x),
                                sigma = sd(x)))),
                      error = function(e) NULL)
fit_bur   <- tryCatch(fitdist(x, "burr",
                              start   = list(shape1 = 2, shape2 = 1,
                                             scale  = median(x) * 0.5),
                              method  = "mle", lower = c(0.01, 0.01, 0),
                              control = list(maxit = 5000, reltol = 1e-10)),
                      error = function(e) NULL)
fit_fre   <- tryCatch(fitdist(x, "frechet",
                              start  = list(shape = 1/0.7, scale = median(x)),
                              method = "mle", lower = c(0.01, 0)),
                      error = function(e) NULL)
get_theo_q <- function(p, distname, fit) {
  switch(distname,
    Normal   = qnorm(p,    mean = fit$estimate["mean"], sd = fit$estimate["sd"]),
    Lognormal= qlnorm(p,   meanlog = fit$estimate["meanlog"],
    sdlog = fit$estimate["sdlog"]),
    Loggamma = qlnorm(p,   meanlog = mean(log(x)), sdlog = sd(log(x))),
    Pareto   = qpareto(p,  shape = fit$estimate["shape"],
    scale = fit$estimate["scale"]),
    Cauchy   = qcauchy(p,  location = fit$estimate["location"],
    scale = fit$estimate["scale"]),
    Student  = qt_ls(p,    df = fit$estimate["df"], mu = fit$estimate["mu"],
                           sigma = fit$estimate["sigma"]),
    Burr     = qburr(p,    shape1 = fit$estimate["shape1"],
    shape2 = fit$estimate["shape2"],
                           scale = fit$estimate["scale"]),
    Frechet  = qfrechet(p, shape = fit$estimate["shape"],
    scale = fit$estimate["scale"])
  )
}
dists <- list(
  list(name = "Normal",    fit = fit_norm),
  list(name = "Lognormal", fit = fit_ln),
  list(name = "Loggamma",  fit = fit_lg),
  list(name = "Pareto",    fit = fit_par),
  list(name = "Cauchy",    fit = fit_cau),
  list(name = "Student",   fit = fit_stu),
  list(name = "Burr",      fit = fit_bur),
```

```
  list(name = "Frechet",   fit = fit_fre)
)
cat("===== YEAR: 2019 =====\n")
cat(sprintf("%-12s  %10s  %10s  %10s  %10s\n",
            "Distrib.", "Q(90%)emp", "Q(90%)teo", "gap90%",
                        "Q(99%)emp"))
cat(sprintf("%-12s  %10s  %10s  %10s  %10s  %10s  %10s\n",
            "", "", "", "", "Q(99%)teo", "gap99%", ""))

cat(strrep("-", 80), "\n")
q90_emp <- as.numeric(quantile(x, 0.90))
q99_emp <- as.numeric(quantile(x, 0.99))
for (d in dists) {
  if (is.null(d$fit)) next
  q90_theo <- tryCatch(get_theo_q(0.90, d$name, d$fit), error = function(e) NA)
  q99_theo <- tryCatch(get_theo_q(0.99, d$name, d$fit), error = function(e) NA)
  gap90 <- abs(q90_emp - q90_theo) / q90_emp * 100
  gap99 <- abs(q99_emp - q99_theo) / q99_emp * 100
  cat(sprintf("%-12s  Q(90%%): emp=%-10.0f theo=%-10.0f gap=%-6.1f%%
  |  Q(99%%): emp=%-10.0f theo=%-10.0f gap=%-6.1f%%\n",
              d$name, q90_emp, q90_theo, gap90, q99_emp, q99_theo, gap99))
}
```

B.4 Extreme Value Analysis : Temporal Stability of the Tail Index 2010–2019)

Hill plots and estimating the Tail Index

```
# Temporal stability of the tail index (2010-2019)
library(evt0)
library(ReIns)
library(tea)
years <- 2010:2019
lk <- c(190, 190, 260, 260, 270, 260, 220, 220, 160, 200)
uk <- c(240, 240, 330, 350, 350, 380, 300, 260, 240, 270)
gamma_year <- numeric(length(years))
k_opt_year <- numeric(length(years))
for (i in seq_along(years)) {
  yr      <- years[i]
  x_yr    <- df[[paste0("Pop", yr)]]
  x_yr    <- x_yr[!is.na(x_yr) & x_yr > 0]
  set.seed(123)
  x_yr_j  <- x_yr + runif(length(x_yr), 0, 0.05)
  kmax_yr <- length(x_yr_j) - 1
  kval_yr <- 1:kmax_yr
  hill_yr    <- as.numeric(mop(x_yr_j, kval_yr, 0, "MOP")$EVI)
  hill_br_yr <- as.numeric(mop(x_yr_j, kval_yr, 0, "RBMOP")$EVI)
  evi_lb_yr  <- hill_yr * (1 - 1.96 / sqrt(kval_yr))
  evi_ub_yr  <- hill_yr * (1 + 1.96 / sqrt(kval_yr))
  k.opt_yr    <- uk[i]
  hill.opt    <- hill_yr[uk[i]]
  hill.br.opt <- hill_br_yr[uk[i]]
  k_opt_year[i] <- k.opt_yr
  gamma_year[i] <- hill.opt
  # Save Hill plot
  png(filename = paste0("Hillplot_", yr, ".png"),
      width = 900, height = 600, res = 150)
  plot(kval_yr, hill_yr, type = "l", col = "grey", lwd = 2,
       main = paste("Hill -", yr),
       xlab = "k", ylab = expression(hat(gamma)),
       xlim = c(0, 500), ylim = c(0, 2))
  lines(kval_yr, hill_br_yr, col = "red",   lwd = 2)
  lines(kval_yr, evi_lb_yr,  col = "black", lty = 2)
  lines(kval_yr, evi_ub_yr,  col = "black", lty = 2)
  abline(h = 0,          col = "darkgreen")
  abline(v = lk[i],      col = "blue", lty = 3, lwd = 1)
  abline(v = uk[i],      col = "blue", lty = 3, lwd = 1)
  abline(h = hill.opt,   col = "red",  lty = 3, lwd = 1)
  legend("topleft", cex = 0.6,
```

```
        legend = c("Hill", "BR-Hill", "95% CI", "Stable region"),
        col    = c("grey", "red", "black", "blue"),
        lty    = c(1, 1, 2, 3), lwd = c(2, 2, 1, 1))
  dev.off()
  cat("Plot saved: Hillplot_", yr, ".png\n", sep = "")
  cat("\n===== Year", yr, "=====\n")
  cat("Stable region: [", lk[i], ",", uk[i], "]\n")
  cat("Hill at k =", uk[i], "      :", round(hill.opt,    4), "\n")
  cat("BR-Hill at k =", uk[i], "   :", round(hill.br.opt, 4), "\n")
  readline(prompt = paste(">>> Plot", yr, "Continue"))
}
# Temporal stability plot
png(filename = "Tailindex_stability.png", width = 900, height = 600, res = 150)
plot(years, gamma_year,
     type = "b", pch = 19, col = "red", lwd = 2,
     xlab = "Year", ylab = expression(hat(gamma)),
     main = expression("Tail index " * hat(gamma) * " by year (2010-2019)"),
     ylim = c(0, max(gamma_year) * 1.4))
abline(h = mean(gamma_year), col = "grey50", lty = 2, lwd = 1.5)
legend("topright",
       legend = c(expression(hat(gamma) * " per year"),
                  paste0("Average = ", round(mean(gamma_year), 3))),
       col = c("red", "grey50"), lty = c(1, 2), pch = c(19, NA))
dev.off()
cat("Plot saved: Tailindex_stability.png\n")
cat("\nSummary across years:\n")
print(data.frame(Year  = years,
                 k_opt = k_opt_year,
                 gamma = round(gamma_year, 4)))
cat("Mean gamma:", round(mean(gamma_year), 4), "\n")
cat("SD gamma:  ", round(sd(gamma_year),   4), "\n")
```

Diagnostic Plots

```
# Exponential QQ-plot + Pareto QQ-plot
for (i in seq_along(years)) {
  yr     <- years[i]
  x_yr   <- df[[paste0("Pop", yr)]]
  x_yr   <- x_yr[!is.na(x_yr) & x_yr > 0]
  set.seed(123)
  x_yr_j <- x_yr + runif(length(x_yr), 0, 0.05)
  kmax_yr <- length(x_yr_j) - 1
  kval_yr <- 1:kmax_yr
  hill_yr  <- as.numeric(mop(x_yr_j, kval_yr, 0, "MOP")$EVI)
  k.opt_yr <- uk[i]
```

```
  hill.opt <- hill_yr[uk[i]]
  sorted_yr <- sort(x_yr_j, decreasing = TRUE)
  kn        <- 1:k.opt_yr
  Ei        <- kn * log(sorted_yr[kn] / sorted_yr[kn + 1])
  theoEi    <- hill.opt * log((k.opt_yr + 1) / (k.opt_yr + 1 - kn))
  #Save plots
  png(filename = paste0("QQplot_", yr, ".png"),
      width = 1200, height = 600, res = 150)
  par(mfrow = c(1, 2))
  # Exponential QQ-plot
  plot(sort(Ei), sort(theoEi),
       main = paste("Exp QQ-plot -", yr),
       xlab = "Empirical quantiles",
       ylab = "Theoretical quantiles",
       pch = 16, cex = 0.6, col = "steelblue")
  lines(sort(theoEi), sort(theoEi), col = "red", lwd = 2)
  # Pareto QQ-plot
  exc <- (sorted_yr / sorted_yr[k.opt_yr])[1:k.opt_yr]
  ParetoQQ(exc[exc > 0],
           main = paste("Pareto QQ-plot -", yr))
  abline(a = 0, b = hill.opt, col = "red", lwd = 2)
  par(mfrow = c(1, 1))
  dev.off()
  cat("Plot saved: QQplot_", yr, ".png\n", sep = "")
  readline(prompt = paste(">>> Plots", yr, "Continue"))
}
```

Weissman estimator for extreme quantile

```
for (i in seq_along(years)) {
  yr     <- years[i]
  x_yr   <- df[[paste0("Pop", yr)]]
  x_yr   <- x_yr[!is.na(x_yr) & x_yr > 0]
  n_yr   <- length(x_yr)
  set.seed(123)
  x_yr_j <- x_yr + runif(n_yr, 0, 0.05)
  kmax_yr <- length(x_yr_j) - 1
  kval_yr <- 1:kmax_yr
  hill_yr  <- as.numeric(mop(x_yr_j, kval_yr, 0, "MOP")$EVI)
  k.opt_yr <- uk[i]
  hill.opt <- hill_yr[k.opt_yr]
  betan    <- 1 - 1 / n_yr
  emp_q <- quantile(x_yr_j, betan, type = 3)
  weiss <- quantile(x_yr_j, 1 - kval_yr / n_yr, type = 3) *
           ((1 - betan) / (kval_yr / n_yr))^(-hill_yr)
```

```
  q.lb <- weiss * exp(-1.96 * log((kval_yr / n_yr) / (1 - betan)) *
                      sqrt(1 / kval_yr) * hill_yr)
  q.ub <- weiss * exp(1.96 * log((kval_yr / n_yr) / (1 - betan)) *
                      sqrt(1 / kval_yr) * hill_yr)
  weiss.opt <- as.numeric(weiss[k.opt_yr])
  png(filename = paste0("Weissman_plot_", yr, ".png"),
      width = 1200, height = 800, res = 150)
  plot(kval_yr, weiss / 1e6,
       type = "l", col = "grey", lwd = 2,
       ylab = "Estimates (millions)", xlab = "k",
       main = paste("Weissman plot -", yr),
       xlim = c(0, 500),
       ylim = c(0, max(weiss.opt / 1e6, na.rm = TRUE) * 3))
  lines(kval_yr, q.lb / 1e6, col = "black", lty = 2)
  lines(kval_yr, q.ub / 1e6, col = "black", lty = 2)
  abline(v = lk[i],             col = "blue",      lty = 3, lwd = 1)
  abline(v = uk[i],             col = "blue",      lty = 3, lwd = 1)
  abline(h = weiss.opt / 1e6,   col = "red",       lty = 3, lwd = 1)
  abline(h = max(x_yr_j) / 1e6, col = "darkgreen", lty = 2, lwd = 1)
  abline(h = emp_q / 1e6,       col = "purple",    lty = 2, lwd = 1)
  legend("topright", cex = 0.7,
         legend = c("Weissman", "95% CI", "Selected estimate",
                    "Sample max", "Empirical q(beta)"),
         col    = c("grey", "black", "red", "darkgreen", "purple"),
         lty    = c(1, 2, 3, 2, 2),
         lwd    = c(2, 1, 1, 1, 1))
  dev.off()
  cat("\n===== Weissman Results -", yr, "=====\n")
  cat("beta = 1-1/n =", round(betan, 6), "\n")
  cat("Hill estimate at k =", k.opt_yr, ":", round(hill.opt, 4), "\n")
  cat("Empirical q(beta):        ",
      round(emp_q / 1e6, 2), "million\n")
  cat("Weissman Q(1-1/n) at k =", k.opt_yr, ":",
      round(weiss.opt / 1e6, 2), "million\n")
  cat("Sample maximum:           ",
      round(max(x_yr_j) / 1e6, 2), "million\n")
  cat("Weissman > Empirical q:   ",
      weiss.opt > emp_q, "\n")
  readline(prompt = paste(">>> Plot", yr, "Continue"))
}
```

B.5 Extreme Value Anaylsis : Spatial Heterogenity

Preparation

```
# Build the 2019 cross-section
x2019 <- df$Pop2019
x2019 <- x2019[!is.na(x2019) & x2019 > 0]
n2019 <- length(x2019)
cat("n (2019 cross-section):", n2019, "\n")
# map cities to US Census Bureau geographic regions
df <- df %>%
  mutate(
    State = gsub(".*,\\s*", "", City),
    Region = case_when(
      State %in% c("Maine", "New Hampshire", "Vermont", "Massachusetts",
                   "Rhode Island", "Connecticut", "New York", "New Jersey",
                   "Pennsylvania") ~ "Northeast",
      State %in% c("Ohio", "Michigan", "Indiana", "Illinois", "Wisconsin",
                   "Minnesota", "Iowa", "Missouri", "North Dakota",
                   "South Dakota", "Nebraska", "Kansas") ~ "Midwest",
      State %in% c("Delaware", "Maryland", "Virginia", "West Virginia",
                   "North Carolina", "South Carolina", "Georgia", "Florida",
                   "Kentucky", "Tennessee", "Alabama", "Mississippi",
                   "Arkansas", "Louisiana", "Oklahoma", "Texas",
                   "District of Columbia") ~ "South",
      State %in% c("Montana", "Idaho", "Wyoming", "Colorado", "New Mexico",
                   "Arizona", "Utah", "Nevada", "Washington", "Oregon",
                   "California", "Alaska", "Hawaii") ~ "West",
      TRUE ~ "Other"
    )
  )
table(df$Region)
```

Full Sample EVT - 2019 Baseline

```
set.seed(123)
x2019_j  <- x2019 + runif(n2019, 0, 0.05)
kmax2019 <- length(which(x2019_j > 0)) - 1
kval2019 <- 1:kmax2019
hill2019    <- as.numeric(mop(x2019_j[x2019_j > 0], kval2019, 0, "MOP")$EVI)
hill2019_br <- as.numeric(mop(x2019_j[x2019_j > 0], kval2019, 0, "RBMOP")$EVI)
evi_lb2019  <- hill2019 * (1 - 1.96 / sqrt(kval2019))
evi_ub2019  <- hill2019 * (1 + 1.96 / sqrt(kval2019))
# We adjust after visual inspection
###############
```

```
lk2019 <- 57
uk2019 <- 150
#################
k_opt2019       <- lk2019
hill_opt2019    <- hill2019[k_opt2019]
hill_br_opt2019 <- hill2019_br[k_opt2019]
png("figures/hill_2019.png", width = 1200, height = 800, res = 150)
plot(kval2019, hill2019, type = "l", col = "grey", lwd = 2,
     xlab = "k", ylab = expression(hat(gamma)),
     main = "Hill plot -- 2019 cross-section (full sample)",
     xlim = c(0, 500), ylim = c(0, 2))
lines(kval2019, hill2019_br, col = "red",   lwd = 2)
lines(kval2019, evi_lb2019,  col = "black", lty = 2)
lines(kval2019, evi_ub2019,  col = "black", lty = 2)
abline(h = 0,               col = "darkgreen")
abline(v = lk2019,          col = "blue", lty = 3)
abline(v = uk2019,          col = "blue", lty = 3)
abline(h = hill_br_opt2019, col = "red",  lty = 3)
legend("topright",
       legend = c("Hill", "BR-Hill", "95% CI", "Stable region",
                  paste0("BR-Hill=", round(hill_br_opt2019, 3))),
       col = c("grey", "red", "black", "blue", "red"),
       lty = c(1, 1, 2, 3, 3), lwd = c(2, 2, 1, 1, 1))
dev.off()
```

Hill Estimator by Geographic Region

```
# We adjust manually (based on the plot)
####################
lk_reg <- c(Northeast = 17, Midwest = 30, South = 23, West = 20)
uk_reg <- c(Northeast = 23, Midwest = 50, South = 38, West = 40)
####################
regions         <- sort(unique(df$Region[df$Region != "Other"]))
gamma_by_region <- data.frame()
png("figures/hill_regions.png", width = 1400, height = 1000, res = 150)
par(mfrow = c(2, 2))
set.seed(123)
for (reg in regions) {
  sub <- df$Pop2019[df$Region == reg]
  sub <- sub[!is.na(sub) & sub > 0]
  if (length(sub) < 20) next
  n_reg  <- length(sub)
  sub_j  <- sub + runif(n_reg, 0, 0.05)
  kmax_s <- length(which(sub_j > 0)) - 1
```

```
kval_s <- 1:kmax_s
hill_s    <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "MOP")$EVI)
hill_s_br <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "RBMOP")$EVI)
evi_lb_s  <- hill_s_br * (1 - 1.96 / sqrt(kval_s))
evi_ub_s  <- hill_s_br * (1 + 1.96 / sqrt(kval_s))
lk_s          <- lk_reg[reg]
uk_s          <- uk_reg[reg]
k_opt_s       <- lk_s
hill_opt_s    <- hill_s[k_opt_s]
hill_br_opt_s <- hill_s_br[k_opt_s]
plot(kval_s, hill_s, type = "l", col = "grey", lwd = 2,
     main = paste0("Hill - ", reg, "  (n=", n_reg, ")"),
     xlab = "k", ylab = expression(hat(gamma)),
     ylim = c(0, 2))
lines(kval_s, hill_s_br, col = "red",   lwd = 2)
lines(kval_s, evi_lb_s,  col = "black", lty = 2)
lines(kval_s, evi_ub_s,  col = "black", lty = 2)
abline(h = 0,               col = "darkgreen")
abline(v = lk_s,            col = "blue", lty = 3, lwd = 1.5)
abline(v = uk_s,            col = "blue", lty = 3, lwd = 1.5)
abline(h = hill_br_opt_s,   col = "red",  lty = 3, lwd = 1)
abline(h = hill_br_opt2019, col = "navy", lty = 2, lwd = 2)
legend("topright", cex = 0.6,
       legend = c("Hill", "BR-Hill", "95% CI", "Stable region",
                  paste0("BR-Hill=", round(hill_br_opt_s,   3)),
                  paste0("Baseline=", round(hill_br_opt2019, 3))),
       col = c("grey", "red", "black", "blue", "red", "navy"),
       lty = c(1, 1, 2, 3, 3, 2), lwd = c(2, 2, 1, 1.5, 1, 2))
cat(sprintf("\nRegion: %s \n", reg))
cat(sprintf("n = %d\n", n_reg))
cat(sprintf("Stable region: [%d, %d]\n", lk_s, uk_s))
cat(sprintf("k_opt = %d  (k_opt/n = %.1f%%)\n", k_opt_s, 100*k_opt_s/n_reg))
cat(sprintf("Hill at k_opt:    %.4f\n", hill_opt_s))
cat(sprintf("BR-Hill at k_opt: %.4f\n", hill_br_opt_s))
gamma_by_region <- rbind(gamma_by_region,
                         data.frame(
                           Region    = reg,
                           n         = n_reg,
                           k_stable  = paste0("[", lk_s, ", ", uk_s, "]"),
                           k_opt     = k_opt_s,
                           k_opt_pct = paste0(round(100*k_opt_s/n_reg, 1),
                           "%"),
                           Hill      = round(hill_opt_s,    4),
                           BR_Hill   = round(hill_br_opt_s, 4)
```

```
                            ))
}
par(mfrow = c(1, 1))
dev.off()
rownames(gamma_by_region) <- NULL
```

LaTeX Table & Tail Index by Region

```
gamma_by_region$k_opt_pct <- gsub("%", "\\\\%", gamma_by_region$k_opt_pct)
# Full-sample row
full_row <- data.frame(
  Region    = "Full sample",
  n         = n2019,
  k_stable  = paste0("[", lk2019, ", ", uk2019, "]"),
  k_opt     = k_opt2019,
  k_opt_pct = paste0(round(100 * k_opt2019 / n2019, 1), "\\%"),
  Hill      = round(hill_opt2019,    4),
  BR_Hill   = round(hill_br_opt2019, 4)
)
tab <- rbind(gamma_by_region, full_row)
rows_tex <- apply(tab, 1, function(r) {
  prefix <- if (r["Region"] == "Full sample") "\\textbf{Full sample}"
  else r["Region"]
  sprintf("  %s & %s & %s & %s & %s & %s & %s \\\\",
          prefix, r["n"], r["k_stable"], r["k_opt"],
          r["k_opt_pct"], r["Hill"], r["BR_Hill"])
})
rows_tex <- c(rows_tex[-length(rows_tex)], "\\midrule",
rows_tex[length(rows_tex)])
latex_table <- c(
  "\\begin{table}[H]",
  "\\centering",
  "\\caption{Tail index estimates by US Census region (2019 cross-section)}",
  "\\label{tab:spatial_hill}",
  "\\small",
  "\\begin{tabular}{lccccc}",
  "\\toprule",
  "\\textbf{Region} & $n$ & \\textbf{Stable region} & $k^*$ & $k^*/n$ &
  \\textbf{Hill} & \\textbf{BR-Hill} \\\\",
  "\\midrule",
  rows_tex,
  "\\bottomrule",
  "\\end{tabular}",
  "\\end{table}"
```

```
)
```

QQ-Plots by regions

```
# Exponential QQ-plots
png("figures/exp_qq_regions.png", width = 1400, height = 1000, res = 150)
par(mfrow = c(2, 2))
set.seed(123)
for (reg in regions) {
  sub <- df$Pop2019[df$Region == reg]
  sub <- sub[!is.na(sub) & sub > 0]
  if (length(sub) < 20) next
  n_reg  <- length(sub)
  sub_j  <- sub + runif(n_reg, 0, 0.05)
  kmax_s <- length(which(sub_j > 0)) - 1
  kval_s <- 1:kmax_s
  hill_s     <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "MOP")$EVI)
  k_opt_s    <- lk_reg[reg]
  hill_opt_s <- hill_s[k_opt_s]
  sorted_s <- sort(sub_j, decreasing = TRUE)
  kn       <- 1:k_opt_s
  Ei       <- kn * log(sorted_s[1:k_opt_s] / sorted_s[2:(k_opt_s + 1)])
  theoEi   <- hill_opt_s * log((k_opt_s + 1) / (k_opt_s + 1 - kn))
  plot(sort(Ei), sort(theoEi),
       main = paste0("Exp QQ - ", reg,
                     "  k=", k_opt_s, " (", round(100*k_opt_s/n_reg, 1), "%)"),
       xlab = "Empirical quantiles", ylab = "Theoretical quantiles",
       pch = 16, cex = 0.7, col = "steelblue")
  lines(sort(theoEi), sort(theoEi), col = "red", lwd = 2)
}
par(mfrow = c(1, 1))
dev.off()
# Pareto QQ-plots
png("figures/pareto_qq_regions.png", width = 1400, height = 1000, res = 150)
par(mfrow = c(2, 2))
set.seed(123)
for (reg in regions) {
  sub <- df$Pop2019[df$Region == reg]
  sub <- sub[!is.na(sub) & sub > 0]
  if (length(sub) < 20) next
  n_reg  <- length(sub)
  sub_j  <- sub + runif(n_reg, 0, 0.05)
  kmax_s <- length(which(sub_j > 0)) - 1
  kval_s <- 1:kmax_s
```

```
  hill_s     <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "MOP")$EVI)
  k_opt_s    <- lk_reg[reg]
  hill_opt_s <- hill_s[k_opt_s]
  sorted_s <- sort(sub_j, decreasing = TRUE)
  exc      <- (sorted_s / sorted_s[k_opt_s])[1:k_opt_s]
  ParetoQQ(exc[exc > 0],
           main = paste0("Pareto QQ - ", reg,
                         "  k=", k_opt_s, " (", round(100*k_opt_s/n_reg, 1),
                         "%)"))
  abline(a = 0, b = hill_opt_s, col = "red", lwd = 2)
}
par(mfrow = c(1, 1))
dev.off()
```

Weissman Estimator by Region

```
weiss_by_region <- data.frame()
png("figures/weissman_regions.png", width = 1400, height = 1000, res = 150)
par(mfrow = c(2, 2))
set.seed(123)
for (reg in regions) {
  sub <- df$Pop2019[df$Region == reg]
  sub <- sub[!is.na(sub) & sub > 0]
  if (length(sub) < 20) next
  n_reg  <- length(sub)
  sub_j  <- sub + runif(n_reg, 0, 0.05)
  kmax_s <- length(which(sub_j > 0)) - 1
  kval_s <- 1:kmax_s
  hill_s_br <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "RBMOP")$EVI)
  k_opt_s       <- lk_reg[reg]
  hill_br_opt_s <- hill_s_br[k_opt_s]
  betan_s <- 1 - 1 / n_reg
  weiss_s <- quantile(sub_j, 1 - kval_s / n_reg, type = 3) *
             ((1 - betan_s) / (kval_s / n_reg))^(-hill_s_br)
  q_lb_s  <- weiss_s * exp(-1.96 * log((kval_s / n_reg) / (1 - betan_s)) *
                           sqrt(1 / kval_s) * hill_s_br)
  q_ub_s  <- weiss_s * exp( 1.96 * log((kval_s / n_reg) / (1 - betan_s)) *
                            sqrt(1 / kval_s) * hill_s_br)
  weiss_opt_s <- as.numeric(weiss_s[k_opt_s])
  ci_lo <- weiss_opt_s * exp(-1.96 * log((k_opt_s / n_reg) / (1 - betan_s)) *
                               sqrt(1 / k_opt_s) * hill_br_opt_s)
  ci_hi <- weiss_opt_s * exp( 1.96 * log((k_opt_s / n_reg) / (1 - betan_s)) *
                               sqrt(1 / k_opt_s) * hill_br_opt_s)
  plot(kval_s, weiss_s / 1e6,
```

```
     type = "l", col = "grey", lwd = 2,
     ylab = "Estimates (millions)", xlab = "k",
     main = paste0("Weissman - ", reg, "  (n=", n_reg, ")"),
     xlim = c(0, min(kmax_s, 100)),
     ylim = c(0, max(weiss_opt_s / 1e6, na.rm = TRUE) * 3))
lines(kval_s, q_lb_s / 1e6, col = "black", lty = 2)
lines(kval_s, q_ub_s / 1e6, col = "black", lty = 2)
abline(v = lk_reg[reg],       col = "blue",      lty = 3, lwd = 1)
abline(v = uk_reg[reg],       col = "blue",      lty = 3, lwd = 1)
abline(h = weiss_opt_s / 1e6, col = "red",       lty = 3, lwd = 1)
abline(h = max(sub_j) / 1e6,  col = "darkgreen", lty = 2, lwd = 1)
legend("topright", cex = 0.65,
       legend = c("Weissman (BR-Hill)", "95% CI",
                  "Selected estimate", "Sample max"),
       col = c("grey", "black", "red", "darkgreen"),
       lty = c(1, 2, 3, 2), lwd = c(2, 1, 1, 1))
cat(sprintf("\nWeissman - %s \n", reg))
cat(sprintf("n = %d | k_opt = %d | k_opt/n = %.1f%%\n",
            n_reg, k_opt_s, 100*k_opt_s/n_reg))
cat(sprintf("BR-Hill at k_opt: %.4f\n", hill_br_opt_s))
cat(sprintf("Weissman Q(1-1/n): %.2f M   95%% CI: [%.2f, %.2f]\n",
            weiss_opt_s/1e6, ci_lo/1e6, ci_hi/1e6))
cat(sprintf("Sample maximum:    %.2f M\n", max(sub_j)/1e6))
cat("\nWeissman at different beta levels:\n")
for (b in c(0.99, 0.995, 0.999, 1 - 1/n_reg)) {
  w  <- quantile(sub_j, 1 - k_opt_s / n_reg, type = 3) *
        ((1 - b) / (k_opt_s / n_reg))^(-hill_br_opt_s)
  c1 <- w * exp(-1.96 * log((k_opt_s / n_reg) / (1 - b)) *
                sqrt(1 / k_opt_s) * hill_br_opt_s)
  c2 <- w * exp( 1.96 * log((k_opt_s / n_reg) / (1 - b)) *
                 sqrt(1 / k_opt_s) * hill_br_opt_s)
  cat(sprintf("  Q(%.4f): %.2f M   95%% CI: [%.2f, %.2f]\n",
              b, w/1e6, c1/1e6, c2/1e6))
}
weiss_by_region <- rbind(weiss_by_region,
                         data.frame(
                           Region    = reg,
                           n         = n_reg,
                           k_opt     = k_opt_s,
                           k_opt_pct = paste0(round(100*k_opt_s/n_reg, 1), "%"),
                           BR_Hill   = round(hill_br_opt_s, 4),
                           Weissman  = paste0(round(weiss_opt_s/1e6, 2), " M"),
                           CI_95     = paste0("[", round(ci_lo/1e6, 2),
                                              ", ", round(ci_hi/1e6, 2), "]")
```

```
                            ))
}
par(mfrow = c(1, 1))
dev.off()
rownames(weiss_by_region) <- NULL
```

Weissman Sensitivity to Beta

```
beta_levels <- c(1, 0.999, 0.995, 0.99)
beta_labels <- c("1-1/n", "0.999", "0.995", "0.99")
cols        <- c("grey", "steelblue", "darkorange", "darkgreen")
png("figures/weissman_regions_sensitivity.png",
    width = 1400, height = 1000, res = 150)
par(mfrow = c(2, 2))
set.seed(123)
for (reg in regions) {
  sub <- df$Pop2019[df$Region == reg]
  sub <- sub[!is.na(sub) & sub > 0]
  if (length(sub) < 20) next
  n_reg  <- length(sub)
  sub_j  <- sub + runif(n_reg, 0, 0.05)
  kmax_s <- length(which(sub_j > 0)) - 1
  kval_s <- 1:kmax_s
  hill_s_br <- as.numeric(mop(sub_j[sub_j > 0], kval_s, 0, "RBMOP")$EVI)
  k_opt_s <- lk_reg[reg]
  uk_s    <- uk_reg[reg]
  weiss_list <- list()
  opt_list   <- c()
  for (i in seq_along(beta_levels)) {
    b <- ifelse(beta_levels[i] == 1, 1 - 1/n_reg, beta_levels[i])
    w <- quantile(sub_j, 1 - kval_s / n_reg, type = 3) *
         ((1 - b) / (kval_s / n_reg))^(-hill_s_br)
    weiss_list[[i]] <- w
    opt_list[i]     <- as.numeric(w[k_opt_s])
  }
  ymax <- min(opt_list[2] * 3, max(sub_j) / 1e6 * 1.5)
  plot(kval_s, weiss_list[[1]] / 1e6,
       type = "n",
       xlim = c(0, min(kmax_s, 100)),
       ylim = c(0, ymax),
       xlab = "k", ylab = "Estimates (millions)",
       main = paste0("Weissman sensitivity - ", reg, "  (n=", n_reg, ")"))
  for (i in seq_along(beta_levels)) {
    lines(kval_s, weiss_list[[i]] / 1e6, col = cols[i], lwd = 2)
    abline(h = opt_list[i] / 1e6, col = cols[i], lty = 3, lwd = 1)
```

```
  }
  abline(v = k_opt_s,           col = "blue",  lty = 3, lwd = 1.5)
  abline(v = uk_s,              col = "blue",  lty = 3, lwd = 1.5)
  abline(h = max(sub_j) / 1e6,  col = "black", lty = 2, lwd = 1.5)
  legend("topright", cex = 0.6,
         legend = c(paste0("beta=", beta_labels),
                    "Selected k*", "Sample max", "Stable region"),
         col = c(cols, "black", "black", "blue"),
         lty = c(1, 1, 1, 1, 3, 2, 3),
         lwd = c(2, 2, 2, 2, 1, 1.5, 1.5))
  cat(sprintf("\nWeissman sensitivity - %s \n", reg))
  cat(sprintf("Sample max: %.2f M\n", max(sub_j) / 1e6))
  for (i in seq_along(beta_levels)) {
    cat(sprintf("  beta=%-6s : %.2f M\n", beta_labels[i], opt_list[i] / 1e6))
  }
}
par(mfrow = c(1, 1))
dev.off()
```

B.6 Predicting City Sizes in 2040

Log-linear Trend Estimation and 2040 Predictions

```
years_vec <- 2010:2019
pop_cols  <- paste0("Pop", years_vec)
df_long <- df %>%
  dplyr::select(City, all_of(pop_cols)) %>%
  pivot_longer(cols = starts_with("Pop"),
               names_to = "Year", values_to = "Population") %>%
  mutate(Year       = as.integer(gsub("Pop", "", Year)),
         Population = as.numeric(Population))
pred2040 <- df_long %>%
  group_by(City) %>%
  summarise(
    n_obs = n(),
    pred_2040   = exp(predict(lm(log(Population) ~ Year),
                              newdata = data.frame(Year = 2040))),
    growth_rate = coef(lm(log(Population) ~ Year))[2],
    .groups = "drop"
  )
top5_grow <- pred2040 %>% arrange(desc(growth_rate)) %>% head(5) %>%
  dplyr::select(City, `Growth rate` = growth_rate,
                `Predicted 2040` = pred_2040) %>%
  mutate(`Growth rate` = round(`Growth rate`, 5),
         `Predicted 2040` = format(round(`Predicted 2040`), big.mark = ","))
top5_decl <- pred2040 %>% arrange(growth_rate) %>% head(5) %>%
  dplyr::select(City, `Growth rate` = growth_rate,
                `Predicted 2040` = pred_2040) %>%
  mutate(`Growth rate` = round(`Growth rate`, 5),
         `Predicted 2040` = format(round(`Predicted 2040`), big.mark = ","))
knitr::kable(top5_grow,
             caption = "Top 5 fastest growing cities (log-linear trend)")
knitr::kable(top5_decl,
             caption = "Top 5 fastest declining cities (log-linear trend)")
```

Heavy Tail Confirmation

```
x2040 <- pred2040$pred_2040
n2040 <- length(x2040)
par(mfrow = c(1, 2))
hist(x2040, nclass = 50, col = "lightblue", border = "white",
     main = "Predicted population 2040", xlab = "Population")
rug(x2040, col = "magenta")
hist(log(x2040), nclass = 40, col = "dodgerblue", border = "white",
     main = "log scale", xlab = "log(Population)")
```

```
rug(log(x2040), col = "magenta")
par(mfrow = c(1, 1))
mom2040 <- Moment(x2040[x2040 > 0])
plot(mom2040$gamma, type = "l", lwd = 1.5, col = "steelblue",
     ylim = c(0, 2), xlim = c(0, 500),
     xlab = "k", ylab = expression(hat(gamma)),
     main = "Moment estimator - heavy tail check (2040 predictions)")
abline(h = 0, col = "magenta", lty = 1, lwd = 2)
```

Hill Estimator

```
set.seed(123)
x2040_j  <- x2040 + runif(n2040, 0, 0.05)
kmax2040 <- length(x2040_j) - 1
kval2040 <- 1:kmax2040

hill2040    <- as.numeric(mop(x2040_j, kval2040, 0, "MOP")$EVI)
hill2040_br <- as.numeric(mop(x2040_j, kval2040, 0, "RBMOP")$EVI)
evi_lb2040  <- hill2040 * (1 - 1.96 / sqrt(kval2040))
evi_ub2040  <- hill2040 * (1 + 1.96 / sqrt(kval2040))
lk2040    <- 60
uk2040    <- 110
k_opt2040 <- uk2040
hill_opt2040    <- hill2040[k_opt2040]
hill_br_opt2040 <- hill2040_br[k_opt2040]
damse2040      <- dAMSE(x2040_j)
k_auto2040     <- damse2040$k0
gamma_auto2040 <- 1 / damse2040$tail.index
plot(kval2040, hill2040, type = "l", col = "grey", lwd = 2,
     main = "Hill - 2040 predicted distribution",
     xlab = "k", ylab = expression(hat(gamma)),
     xlim = c(0, 500), ylim = c(0, 2))
lines(kval2040, hill2040_br, col = "red",   lwd = 2)
lines(kval2040, evi_lb2040,  col = "black", lty = 2)
lines(kval2040, evi_ub2040,  col = "black", lty = 2)
abline(h = 0,            col = "darkgreen")
abline(v = lk2040,       col = "blue", lty = 3, lwd = 1)
abline(v = uk2040,       col = "blue", lty = 3, lwd = 1)
abline(h = hill_opt2040, col = "red",  lty = 3, lwd = 1)
legend("topleft", cex = 0.6,
       legend = c("Hill", "BR-Hill", "95% CI", "Stable region"),
       col    = c("grey", "red", "black", "blue"),
       lty    = c(1, 1, 2, 3), lwd = c(2, 2, 1, 1))
cat("\n===== 2040 Predicted Distribution =====\n")
cat("Stable region: [", lk2040, ",", uk2040, "]\n")
```

```
cat("Hill at k =", k_opt2040, "      :", round(hill_opt2040,    4), "\n")
cat("BR-Hill at k =", k_opt2040, "   :", round(hill_br_opt2040, 4), "\n")
cat("dAMSE k0              :", k_auto2040,              "\n")
cat("dAMSE gamma           :", round(gamma_auto2040,   4), "\n")
```

QQ-Plots

```
sorted2040 <- sort(x2040_j, decreasing = TRUE)
kn2040     <- 1:uk2040
Ei2040     <- kn2040 * log(sorted2040[kn2040] / sorted2040[kn2040 + 1])
theoEi2040 <- hill_opt2040 * log((uk2040 + 1) / (uk2040 + 1 - kn2040))
par(mfrow = c(1, 2))
plot(sort(Ei2040), sort(theoEi2040),
     main = "Exp QQ-plot - 2040 predictions",
     xlab = "Empirical quantiles",
     ylab = "Theoretical quantiles",
     pch = 16, cex = 0.6, col = "steelblue")
lines(sort(theoEi2040), sort(theoEi2040), col = "red", lwd = 2)
exc2040 <- (sorted2040 / sorted2040[uk2040])[1:uk2040]
ParetoQQ(exc2040[exc2040 > 0],
         main = "Pareto QQ-plot - 2040 predictions")
abline(a = 0, b = hill_opt2040, col = "red", lwd = 2)
par(mfrow = c(1, 1))
```

Weissman Estimator

```
# Weissman estimator 2040 predicted distribution
betan2040 <- 1 - 1 / n2040
emp_q2040 <- quantile(x2040_j, betan2040, type = 3)
weiss2040 <- quantile(x2040_j, 1 - kval2040 / n2040, type = 3) *
             ((1 - betan2040) / (kval2040 / n2040))^(-hill2040)
q.lb2040 <- weiss2040 * exp(-1.96 * log((kval2040 / n2040) / (1 - betan2040)) *
                            sqrt(1 / kval2040) * hill2040)
q.ub2040 <- weiss2040 * exp( 1.96 * log((kval2040 / n2040) / (1 - betan2040)) *
                           sqrt(1 / kval2040) * hill2040)
weiss.opt2040 <- as.numeric(weiss2040[k_opt2040])
plot(kval2040, weiss2040 / 1e6,
     type = "l", col = "grey", lwd = 2,
     ylab = "Estimates (millions)", xlab = "k",
     main = "Weissman plot - 2040 predictions",
     xlim = c(0, 500),
     ylim = c(0, max(weiss.opt2040 / 1e6, na.rm = TRUE) * 3))
lines(kval2040, q.lb2040 / 1e6, col = "black",  lty = 2)
lines(kval2040, q.ub2040 / 1e6, col = "black",  lty = 2)
abline(v = lk2040,              col = "blue",    lty = 3, lwd = 1)
```

```
abline(v = uk2040,              col = "blue",    lty = 3, lwd = 1)
abline(h = weiss.opt2040 / 1e6, col = "red",     lty = 3, lwd = 1)
abline(h = max(x2040_j) / 1e6,  col = "darkgreen", lty = 2, lwd = 1)
abline(h = emp_q2040 / 1e6,     col = "purple",  lty = 2, lwd = 1)
legend("topright", cex = 0.7,
       legend = c("Weissman", "95% CI", "Selected estimate",
                  "Sample max", "Empirical q(beta)"),
       col    = c("grey", "black", "red", "darkgreen", "purple"),
       lty    = c(1, 2, 3, 2, 2), lwd = c(2, 1, 1, 1, 1))
cat("\n===== Weissman Results - 2040 predictions =====\n")
cat("n cities:                ", n2040, "\n")
cat("k selected (uk):         ", k_opt2040, "\n")
cat("beta = 1-1/n =", round(betan2040, 6), "\n")
cat("Hill estimate at k =", k_opt2040, ":", round(hill_opt2040, 4), "\n")
cat("Empirical q(beta):       ", round(emp_q2040 / 1e6, 2), "million\n")
cat("Weissman Q(1-1/n) at k =", k_opt2040, ":",
    round(weiss.opt2040 / 1e6, 2), "million\n")
cat("Sample maximum:           ",
    round(max(x2040_j) / 1e6, 2), "million\n")
cat("Weissman > Empirical q:  ", weiss.opt2040 > emp_q2040, "\n")
```